# Bridging HSI and LiDAR Data With Frequency-Domain Hierarchical Fusion for Enhanced Classification

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***Abstract*— Remote sensing data from hyperspectral imaging (HSI) and light detection and ranging (LiDAR) provide complementary perspectives for terrain and object analysis. However, existing methods for multimodal data fusion primarily focus on spatial-domain feature alignment, often overlooking the potential of frequency-domain information to enhance classification accuracy. To bridge this gap, we introduce the frequency-domain hierarchical perception fusion network (FHPF-Net), a novel framework for precise classification of remote sensing images. This network leverages both spatial-domain information and frequency-domain information and provides a new perspective for heterogeneous data integration. To extract and utilize frequency-domain features, we propose the high-to-low spectral separation and mining (HLSSM) module, which isolates high-frequency details such as edges and textures from low-frequency structural patterns in HSI and LiDAR data. This separation facilitates targeted feature extraction (FE) while preserving crucial contextual information. Additionally, we introduce the hierarchical superimposed multidomain information fusion (HSMIF) module, which employs a multilevel fusion strategy to integrate spatial- and frequency-domain features, ensuring consistency and complementarity between the two data sources. Finally, we introduce a learnable voting pre-label fusion (LVPF) strategy to effectively integrate multibranch outputs, enhancing classification performance and model robustness. The proposed FHPF-Net effectively captures diverse responses across heterogeneous data types, enabling robust classification in complex environments. Extensive experiments demonstrate that our approach significantly outperforms state-of-the-art methods.**

***Index Terms*— Category-aware, frequency domain, hyperspectral imaging (HSI), light detection and ranging (LiDAR) image.**

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## I. Introduction

REMOTE sensing is a crucial Earth observation technology that utilizes airborne or spaceborne instruments to measure various physical signals from the Earth's surface [1], [2]. Land cover classification is a key application in the field of remote sensing interpretation, aiming to distinguish different types of land cover such as forests, water bodies, urban structures, farmland, and bare soil [3], [4], [5], [6]. This task provides valuable foundational data for resource surveys, environmental monitoring, urban planning, and disaster warning systems [7], [8], [9], [10], [11].

To achieve these goals, various types of remote sensing data demonstrate unique advantages in information acquisition. For instance, hyperspectral images capture the spatial distribution of surface materials with high accuracy while also providing fine-grained spectral details across a broad range of wavelengths—from the visible to the infrared spectrum—thus offering rich spectral and spatial information [12], [13]. However, due to the inherent tradeoff between spectral and spatial resolution, hyperspectral imaging (HSI) sensors often sacrifice spatial resolution to achieve high spectral fidelity [14], [15]. In contrast, light detection and ranging (LiDAR) data acquire spatial information by emitting laser pulses and measuring the returned signals, which include the distance to target surfaces, reflected intensity, and other physical characteristics [16], [17]. This enables LiDAR to capture the detailed geometric properties of objects, such as their shape, structure, and height [9].

In the early stages, handcrafted methods were designed separately for each modality. For HSI, traditional classifiers such as random forest (RF) [18], [19], support vector machine (SVM) [20], [21], and extreme learning machine (ELM) [22] relied on spectral reflectance features. For LiDAR, geometric structure and height statistics were exploited using techniques like conditional random fields [23] or multikernel sparse representation [24]. However, these manual feature extraction FE) methods lack scalability and often underperform in complex scenes. With the rise of deep learning, convolutional neural

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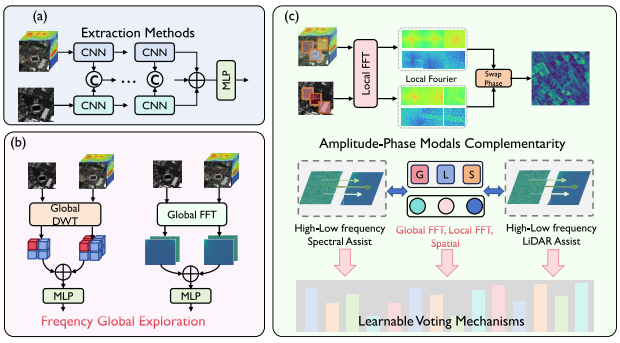



Fig. 1. Comparison of different FE paradigms. (a) Existing spatial-based methods typically extract spatial–spectral features using CNN but often neglect frequency-domain information. (b) Existing frequency-based methods mainly perform global transformations (e.g., DWT or FFT) to obtain holistic frequency cues, yet lack local granularity and adaptive fusion across modalities. (c) Our proposed frequency hierarchical fusion method combines spatial, local, and global frequency features, explicitly models amplitude–phase complementarity between modalities, and employs a learnable voting mechanism to enhance robustness in decision-level fusion.

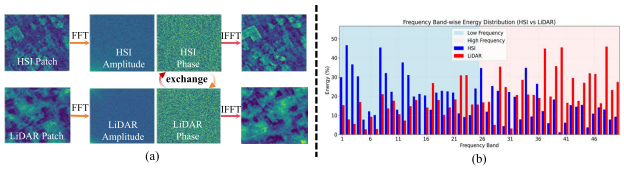



Fig. 2. Motivation. (a) Frequency component decomposition of HSI and LiDAR data. Amplitude and phase are separately visualized, and selective swapping is applied to reconstruct fused frequency features. (b) Band-wise energy distribution comparison between HSI and LiDAR in the frequency domain. HSI concentrates energy in low-frequency bands (blue), while LiDAR exhibits stronger responses in high-frequency regions (red), revealing their complementary characteristics. This supports the necessity of frequency-domain fusion.

networks (CNNs) [25], [26], [27], [28], [29], [30] and Transformer [31], [32], [33], [34] have shown powerful abilities to extract spatial–spectral features from HSI. For example, Lee and Kwon [30] introduced multiscale convolutional filters to capture local spatial–spectral relationships among neighboring pixels. Sun et al. [31] proposed the spectral–spatial feature tokenization Transformer (SSFTT) to capture spectral–spatial and high-level semantic features. However, single-source HSI data classification tasks often encounter misclassification issues in complex environments, such as forests and ocean surfaces, where features may exhibit similar spectral characteristics.

To address this limitation, collaborative methods [35], [36], [37], [38] combining HSI and LiDAR data have emerged as a mainstream trend in feature classification research. By combining the rich spectral details of HSI with the precise elevation and structural cues of LiDAR, these methods aim to compensate for the weaknesses of single-modality systems. As illustrated in Fig. 1, existing approaches can be broadly categorized into two classes.

1) *Spatial-Domain Methods [34], [35], [36], [39], [40]:* For example, Zhao et al. [35] proposed a dual-branch architecture that combines hierarchical CNNs and Transformers to fuse multisource features. Yao et al. [36] introduced position-shared vision Transformers and separable convolutions in parallel branches to enhance cross-modal interaction. While these methods have achieved strong classification results, they focus primarily on spatial feature alignment, often neglecting the intrinsic frequency patterns unique to HSI and LiDAR. This may suppress modality-specific characteristics, such as spectral continuity in HSI or structural abruptness in LiDAR.
2) *Frequency-Domain Methods [38], [41], [42], [43], [44]:* As shown in Fig. 1(b), methods such as FDNet [41] and HFNet [42] utilize global frequency transforms (e.g., fast Fourier transform (FFT) and DWT) to extract modality-specific patterns. Song et al. [43] proposed a height-guided hierarchical separation mechanism to decouple shared and specific features between HSI and LiDAR. Despite these advances, most frequency-domain models are single level and global, lacking the ability to capture local granularity or hierarchical semantics, and often fail to progressively integrate global and local frequency cues.

To further investigate the intrinsic complementarity between HSI and LiDAR data in the frequency domain, we conduct an empirical frequency component analysis, as shown in Fig. 2. In Fig. 2(a), we apply the FFT to decompose each modality into amplitude and phase components, followed by a selective cross-modal exchange and reconstruction. While the amplitude and phase maps do not show visually striking differences, the fused feature reconstructed from exchanged components exhibits significantly enhanced spatial structure and spectral consistency. This indicates that amplitude–phase interactions across modalities serve as a valuable basis for frequency-domain fusion, enabling more coherent joint representations. Fig. 2(b) further quantifies this complementarity by comparing the frequency band-wise energy distributions of HSI and LiDAR. The results show that HSI concentrates energy in low-frequency bands, reflecting its strength in capturing global spectral continuity. In contrast, LiDAR exhibits higher responses in high-frequency regions, highlighting its advantage in representing fine spatial structures such as object boundaries and elevation discontinuities. This difference in spectral energy distribution aligns well with the physical characteristics of the two modalities and reveals their natural frequency-domain complementarity. These observations motivate a shift from conventional spatial-domain fusion strategies toward a frequency-domain perspective, which offers a more structured and interpretable approach for integrating heterogeneous data. Based on this insight, we propose two key research questions: 1) *how to enhance the correlation and expressiveness of spectral–spatial–elevation information from the frequency-domain perspective?* and 2) *how to develop a hierarchical, locality-aware frequency fusion strategy that enables progressive integration of multiscale spectral information, addressing the lack of local granularity in existing global frequency-domain methods?*

To address the aforementioned scientific questions, we propose a frequency-domain hierarchical perception fusion network (FHPF-Net) for precise remote sensing image classification. Specifically, to tackle the first research question, we propose a high-to-low spectral separation and mining (HLSSM) module, which is designed to effectively separate the frequency components of the input HSI and LiDAR data, allowing us to isolate high-frequency details—such as

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fine textures and edges—from low-frequency information that captures broader structural patterns. This separation ensures that both fine-grained details and larger contextual information are effectively utilized in the classification process. To address the second research question, we also propose a hierarchical superimposed multidomain information fusion (HSMIF) module, which emphasizes the complementarity of different data sources in both spatial and frequency domains and incorporates a tightly connected hierarchical stacking learning paradigm to facilitate multilevel fusion of frequency features, ultimately improving classification performance. In addition, we propose an innovative learnable voting pre-label fusion (LVPF) strategy to effectively integrate the final outputs of the three branches. This strategy dynamically assigns learnable weights to each branch's predictions, adjusting their contributions to more accurately determine the final class of each pixel. The LVPF method fully leverages the complementary strengths of the multibranch model, enhancing not only classification accuracy but also the model's adaptability and generalization capabilities in complex scenarios. Our main contributions are summarized as follows.

1) We propose a novel joint classification method for HSI and LiDAR remote sensing images, called FHPF-Net. This method focuses on the extraction and integration of frequency information from heterogeneous data. Through frequency-domain analysis, it is able to capture the frequency features and patterns of the entire scene, thereby significantly enhancing the classification performance of remote sensing images.
2) The HLSSM module is proposed to effectively separate the high- and low-frequency components of HSI and LiDAR data. It extracts unique spectral and elevation features from these heterogeneous data sources across different frequency components, ensuring the integrity of structural patterns and textures.
3) We propose an HSMIF module to effectively integrate the heterogeneous features of HSI and LiDAR data through multilevel hierarchical fusion (spatial and global–local frequency) in spatial and frequency domains. It not only ensures the consistency of different data sources in the two domains but also significantly results in more robust and fine-grained fusion.
4) An LVPF strategy is proposed to dynamically fuse multibranch outputs through learnable weights, effectively leveraging modal's complementarity to enhance classification accuracy and model generalization.

## II. Proposed Methodological Framework

The overall framework of our proposed novel FHPF-Net is illustrated in Fig. 3. This framework consists of four main components: preprocessing, FE, fusion, and classification. Specifically, we utilize PCA for dimensionality reduction as part of the preprocessing step. The FE and fusion section is divided into three branches: spectral frequency FE, elevation frequency FE, and feature fusion. To efficiently capture the rich frequency-domain spectral and elevation information from HSI and LiDAR data, we introduce a novel HLSSM module within the spectral frequency FE branch. In the feature fusion branch, we propose a hierarchical stacking frequency information fusion (HSFIF) module that leverages spatial features alongside both global and local frequency-domain information to ensure the consistency of heterogeneous data. This is accomplished by incorporating a tightly connected hierarchical stacking learning paradigm to facilitate multilevel fusion of frequency features.

### A. Data Preprocessing

HSI captures multiple bands ranging from ultraviolet to near infrared (typically tens to hundreds of bands), providing rich spectral features. However, its large size results in significant computational costs. To address this issue, we employ principal component analysis (PCA) [45] to extract the top $k$ principal components from the HSI data, effectively reducing the spectral dimensionality while preserving essential information for classification.

We define the HSI data and LiDAR data as $X_{\text{hsi}} \in \mathcal{R}^{m\times n\times l}$ and $\mathbf{X}_{\text{lidar}} \in \mathcal{R}^{m\times n}$, where $m$ and $n$ represent the width and height of the heterogeneous data, respectively, and $l$ denotes the spectral bands of the HSI data. Both datasets cover the same area of the ground surface. For each pixel, we can represent it with a one-hot encoded category vector. After applying PCA, the HSI data transform from $X_{\text{hsi}} \in \mathcal{R}^{m\times n\times l}$ to $X_{\text{hsi}}^{\text{pca}} \in \mathcal{R}^{m\times n\times b}$, where $b$ represents the number of retained spectral components.

Next, for both HSI and LiDAR data, we perform 3-D and 2-D patch extraction, resulting in a cubic patch ${X_{\text{hsi}}}^{P} \in \mathcal{R}^{s\times s\times b}$ for the HSI data and a 2-D patch $\mathbf{X}_{\text{lidar}}^{P} \in \mathcal{R}^{s\times s}$ for the LiDAR data, where $s$ indicates the patch size. We use the index of the center pixel to label each patch. To ensure the integrity of edge regions, we apply padding to the edge pixels, with a padding width of $(s-1)/2$. Finally, we eliminate patches labeled as zero and divide the remaining samples into training and testing sets.

### B. HLSSM Module

HSI data reveal multiple-frequency components corresponding to various remote sensing spectral features, aiding in the precise identification and classification of land cover. In the frequency domain, the spectral distribution of HSI data is typically smooth and exhibits continuity across the wavelength range. Additionally, the periodic characteristics of HSI signals in the frequency domain respond significantly to different categories of land cover, such as the varying spectral responses of vegetation and soil.

In contrast, LiDAR data present unique frequency-domain attributes that highlight the spatial frequency characteristics of remote sensing data. Here, high-frequency components correspond to fine details, such as the edges of buildings, while low-frequency components capture overall shapes, like terrain contours. Therefore, we approach effective feature representation from a novel perspective, focusing on extracting the intrinsic properties of these two types of remote sensing images from the frequency domain.

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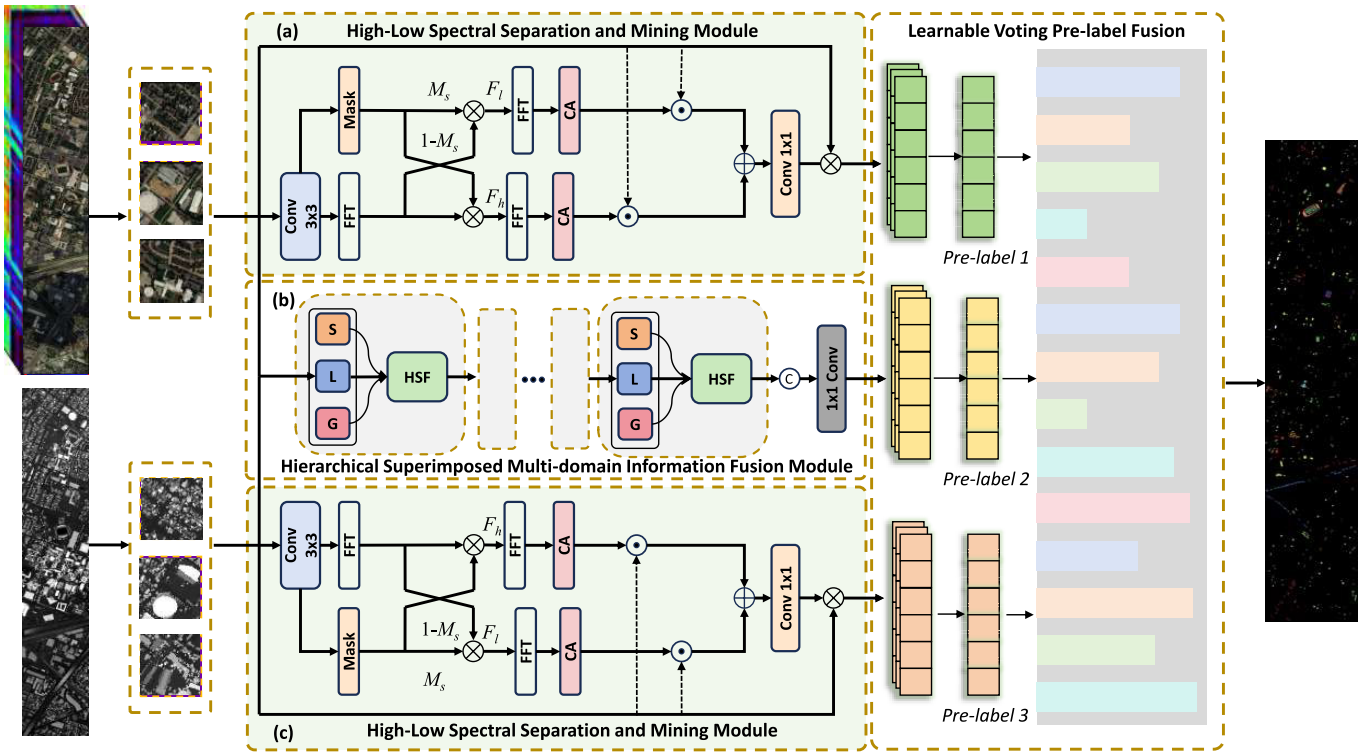



Fig. 3. Overview of the proposed FHPF-Net. The frame work consists of four main components: preprocessing with PCA-based dimensionality reduction; (a) and (c) FE branches for spectral (HSI) and elevation (LiDAR) data processing using the proposed HLSSM module; (b) HSMIF module, which integrates spatial- and frequency-domain features across local and global scales; and LVPF strategy that adaptively aggregates multibranch outputs for final classification.

As shown in Fig. 3, for the given HSI ${X_{\text{hsi}}}^P \in \mathcal{R}^{s\times s\times b}$ and LiDAR data $\mathbf{X}_{\text{lidar}}^P \in \mathcal{R}^{s\times s}$, we aim to extract different frequency representations from heterogeneous data under targeted adaptive decoupling of spectral guidance. The main process is divided into three steps: frequency-domain transformation, mask generation, and feature modulation.

In the frequency-domain transformation step, we apply the FFT [46] to convert the spatial representations of both the HSI and LiDAR data into their respective frequency-domain representations. Specifically, for the HSI patch ${X_{\text{hsi}}}^P \in \mathcal{R}^{s\times s\times b}$, we compute the FFT along the spatial dimensions, resulting in a transformed representation denoted as ${X_{\text{hsi}}}^{\text{freq}} \in \mathcal{R}^{s\times s\times b}$. Similarly, for the LiDAR patch ${\mathbf{X}_{\text{lidar}}}^P \in \mathcal{R}^{s\times s}$, we perform the FFT to obtain its frequency-domain representation ${\mathbf{X}_{\text{lidar}}}^{\text{freq}} \in \mathcal{R}^{s\times s}$. This process can be expressed as

$$X_{\text{hsi}}^{\text{freq}}[u,v,b] = \sum_{m=0}^{s-1}\sum_{n=0}^{s-1} {X_{\text{hsi}}}^P[m,n,b] e^{-j\frac{2\pi}{s}(um+vn)} \tag{1}$$

$$\mathbf{X}_{\text{lidar}}^{\text{freq}}[u,v] = \sum_{m=0}^{s-1}\sum_{n=0}^{s-1} \mathbf{X}_{\text{lidar}}^P[m,n] e^{-j\frac{2\pi}{s}(um+vn)} \tag{2}$$

where $s$ is the size of the patch and $u$ and $v$ are the frequency indices corresponding to the spatial dimensions.

To effectively separate the high- and low-frequency components of the data, we create masks to isolate the low- and high-frequency components from the input tensor $X_{\text{hsi}}^P$ and $\mathbf{X}_{\text{lidar}}^P$. First, we identify the center coordinates of the frequency spectrum, denoted as $\text{c}_h$ and $\text{c}_w$, where $h$ and $w$ are the height and width of the input tensor, respectively. To separate low-/high-frequency components, we create two masks: a low-frequency mask $\mathbf{M}_{\text{low}}$ and a high-frequency mask $\mathbf{M}_{\text{high}}$. We define the low-frequency region size to be $\text{low}_\sigma = (\min(\text{center}_h, \text{center}_w)/m)$, which ensures that the low-frequency mask captures a quarter of the minimum dimension of the frequency spectrum, effectively isolating the significant low-frequency components for analysis. $m$ is the scaling factor for the low-frequency mask. The low-frequency mask $\mathbf{M}_{\text{low}}$ is set to 1 within this defined region centered in the frequency domain and is 0 elsewhere

$$\mathbf{M}_{\text{low}}[i,j] = \begin{cases} 1, & \text{if } |i-\text{c}_h| < \text{low}_\sigma \text{ and } |j-\text{c}_w| < \text{low}_\sigma \\ 0, & \text{otherwise.} \end{cases} \tag{3}$$

The high-frequency mask $M_{\text{high}}$ is then defined as the complement of the low-frequency mask $1-\mathbf{M}_{\text{low}}$. Using these masks, we extract the low-/high-frequency component from the input tensor $X_{\text{hsi}}^{\text{freq}}$ or $\mathbf{X}_{\text{lidar}}^{\text{freq}}$

$$\mathbf{x}_{\text{low}}^{\text{hsi}} = X_{\text{hsi}}^{\text{freq}} \cdot \mathbf{M}_{\text{low}}, \quad \mathbf{x}_{\text{high}}^{\text{hsi}} = X_{\text{hsi}}^{\text{freq}} \cdot \mathbf{M}_{\text{high}} \tag{4}$$

$$\mathbf{x}_{\text{low}}^{\text{lidar}} = \mathbf{X}_{\text{lidar}}^{\text{freq}} \cdot \mathbf{M}_{\text{low}}, \quad \mathbf{x}_{\text{high}}^{\text{lidar}} = \mathbf{X}_{\text{lidar}}^{\text{freq}} \cdot \mathbf{M}_{\text{high}} \tag{5}$$

where $\mathbf{x}_{\text{low}}^{\text{hsi}}$ and $\mathbf{x}_{\text{high}}^{\text{hsi}}$ represent the low-/high-frequency features of the HSI frequency domain, respectively, and $\mathbf{x}_{\text{low}}^{\text{lidar}}$ and $\mathbf{x}_{\text{high}}^{\text{lidar}}$ are the low-/high-frequency features of the LiDAR frequency domain, respectively.

Finally, we perform an inverse Fourier transform to obtain the decoupled adaptive frequency-domain features $\hat{\mathbf{x}}_{\text{low}}^{\text{hsi}}$, $\hat{\mathbf{x}}_{\text{high}}^{\text{hsi}}$, $\hat{\mathbf{x}}_{\text{low}}^{\text{lidar}}$, and $\hat{\mathbf{x}}_{\text{high}}^{\text{lidar}}$. Next, we will emphasize the relevant

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features and suppress redundant ones from the obtained adaptive frequency-domain features through channel modulation. Specifically, we perform an element-wise multiplication of the input features with the emphasized relevant features, followed by an addition operation to fully leverage the unique properties of the different frequency-domain features. Finally, we apply a $1 \times 1$ convolution to generate the final salient features, which are then combined with the original input through element-wise multiplication. The same operations are applied to both HSI and LiDAR data. The process for HSI data can be formally expressed as follows:

$$\bar{\mathbf{x}}_{\text{low}}^{\text{hsi}} = \mathcal{C}\left(\hat{\mathbf{x}}_{\text{low}}^{\text{hsi}}\right) \circ X_{\text{hsi}}^{P}, \quad \bar{\mathbf{x}}_{\text{high}}^{\text{hsi}} = \mathcal{C}\left(\hat{\mathbf{x}}_{\text{high}}^{\text{hsi}}\right) \circ X_{\text{hsi}}^{P} \tag{6}$$

$$F_{\text{salient}} = \text{Conv}_{1\times1}\left(\bar{\mathbf{x}}_{\text{low}}^{\text{hsi}} + \bar{\mathbf{x}}_{\text{high}}^{\text{hsi}}\right) \otimes X_{\text{hsi}}^{P} \tag{7}$$

where $\bar{\mathbf{x}}_{\text{low}}^{\text{hsi}}$ and $\bar{\mathbf{x}}_{\text{high}}^{\text{hsi}}$ indicate the emphasized relevant features after channel modulation; $F_{\text{salient}}$ denotes the output of HLSSM module; $\mathcal{C}(\cdot)$ and $\text{Conv}_{1\times1}$ are the channel attention and the a $1 \times 1$ convolution layer, respectively; and $\circ$ and $\otimes$ denote the dot product and element-wise multiplication operation, respectively.

### C. HSMIF Module

To ensure the consistency and efficient fusion of heterogeneous data features, we propose an HSMIF module, which extracts complementary features from both spatial and frequency domains of heterogeneous data and promotes multisource data fusion through a densely connected hierarchical superposition module. As shown in Fig. 3(b), our HSMIF module comprises four components: spatial extraction, local frequency-domain extraction, global frequency-domain extraction, and an HSF module.

*1) Spatial Extraction:* As shown in Fig. 4, we separately extract local features from HSI and LiDAR data using two $3 \times 3$ convolutions, resulting in $F_s^{\text{hsi}}$ and $F_s^{\text{lidar}}$. Next, we perform a concatenation operation to obtain a combined feature representation, followed by applying multiple ResBlocks [47] and spatial attention to fuse the distinct responses of the two data types for the same region. This process can be expressed as

$$F_s^{\text{hsi}}, F_s^{\text{lidar}} = \mathcal{C}_3\left(X_{\text{hsi}}^{P}\right)\mathcal{C}_3\left(\mathbf{X}_{\text{lidar}}^{P}\right) \tag{8}$$

$$F_s^{m} = \text{Cat}\left[F_s^{\text{hsi}}, F_s^{\text{lidar}}\right] \tag{9}$$

$$F_s = \text{Res}_2\left(\mathcal{S}_2\left(\text{Res}_1\left(\mathcal{S}_1\left(F_s^{m}\right)\right)\right)\right) \tag{10}$$

where $F_s$ is the spatial features; $\mathcal{C}_3$ denotes the $3 \times 3$ convolution layer; and $\mathcal{S}_i$ and $\text{Res}_i$ denote the $i$th spatial attention and residual block, respectively.

*2) Global Frequency Information Extraction:* It is widely known that the Fourier transform enables the conversion of an image from the spatial domain to the frequency domain, where Fourier-based methods are inherently effective at modeling global information. In the Fourier domain, an image's frequency representation typically comprises amplitude and phase components, which describe the strength and location of frequency elements, respectively. Among these, phase information is crucial for capturing an image's structure and shape, which determines the edges and contours in remote sensing imagery. To ensure consistent feature response for relative positioning in HSI and LiDAR data, we perform targeted operations on the phase spectrum.

Specifically, we perform FFT operations on both HSI and LiDAR data to obtain the respective amplitude and phase spectra. Assume that the features of the HSI and LiDAR are $X_{\text{hsi}}^{P}$ and $\mathbf{X}_{\text{lidar}}^{P}$, respectively. Defining the FFT operation as $\mathcal{F}(\cdot)$, this process can be expressed as

$$\mathcal{F}\left(X_{\text{hsi}}^{P}\right) = A_{\text{hsi}} \cdot e^{i\Phi_{\text{hsi}}}, \quad \mathcal{F}\left(\mathbf{X}_{\text{lidar}}^{P}\right) = \mathbf{A}_{\text{lidar}} \cdot e^{i\Phi_{\text{lidar}}} \tag{11}$$

where $A_{\text{hsi}}$ and $\mathbf{A}_{\text{lidar}}$ denote the amplitude spectra of HSI and LiDAR, respectively, capturing the intensity of each frequency component; and $\Phi_{\text{hsi}}$ and $\Phi_{\text{lidar}}$ are the phase spectra, which encode the spatial structure and relative positioning information essential for edge and contour representation in remote sensing imagery.

Next, we concatenate the phase spectrum of the HSI data with the amplitude spectrum of the LiDAR data and vice versa for the LiDAR features. After this, we execute the inverse Fourier transform to convert the global frequency representations back to the spatial domain. Then, we apply a simple ReLU activation function followed by a $1 \times 1$ convolution to each of them. To capture significant responses under the same structure, we perform element-wise multiplication of the corresponding HSI and LiDAR features, which serves as an attention mechanism to obtain the global frequency features $F_g^{\text{hsi}}$ and $F_g^{\text{lidar}}$. The entire process can be expressed with the following formulas:

$$F_g^{\text{hsi}} = \mathcal{R}\left(\mathcal{C}_1\left(\mathcal{F}^{-1}\left(\text{Cat}\left(A_{\text{hsi}}, e^{i\Phi_{\text{lidar}}}\right)\right)\right)\right) \otimes X_{\text{hsi}} \tag{12}$$

$$F_g^{\text{lidar}} = \mathcal{R}\left(\mathcal{C}_1\left(\mathcal{F}^{-1}\left(\text{Cat}\left(\mathbf{A}_{\text{lidar}}, e^{i\Phi_{\text{hsi}}}\right)\right)\right)\right) \otimes \mathbf{X}_{\text{lidar}}^{P} \tag{13}$$

$$F_g = F_{\text{hsi}}^{g} + F_{\text{lidar}}^{g} \tag{14}$$

where $\mathcal{R}$ represents the ReLU activation function and $\mathcal{F}^{-1}$ denotes the inverse Fourier transform.

*3) Local Frequency Information Extraction:* To supplement the local frequency-domain information, we perform overlapping partitioning of the two data sources to extract information from different locations. We utilize the partitioning function $\sigma(\cdot)$ to identify the regions with the highest response values, ensuring that the regions correspond to each other. Specifically, we divide the data into four nonoverlapping regions and use the partitioning function to maintain consistency across these regions. The process can be expressed as follows:

$$R_i = \sigma\left(P_i^{\text{hsi}}, P_i^{\text{lidar}}\right), \quad i = 1, 2, 3, 4 \tag{15}$$

where $R_i$ represents the regions identified based on the highest responses from both HSI and LiDAR features and $\sigma$ represents the partition function. We adopt cosine similarity to achieve this goal.

Next, we perform the same operations on each partition as we did for the global frequency-domain extraction to facilitate the fusion of local information. Finally, we apply the inverse Fourier transform to convert this information back into the spatial domain. Thus, we obtain the local frequency-domain features for the corresponding regions, which can be represented as

$$F_{\text{local}}^{\text{hsi}} = \mathcal{R}\left(\mathcal{C}_1\left(\mathcal{F}^{-1}\left(\text{Cat}\left(A_{\text{hsi}}^{i}, e_i^{i\Phi_{\text{lidar}}}\right)\right)\right)\right) \tag{16}$$

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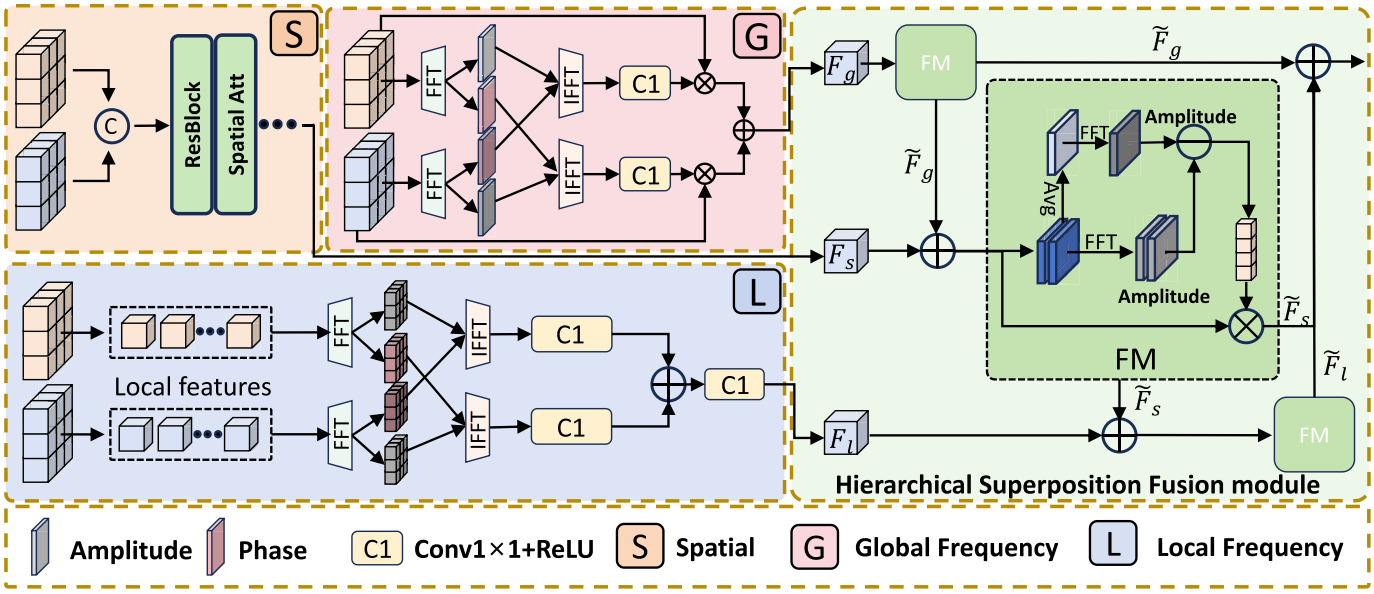



Fig. 4. Pipeline of the proposed HSMIF module. The module consists of four key components: 1) spatial FE using convolution and spatial attention; 2) global frequency FE via Fourier transform and phase–amplitude interaction between HSI and LiDAR modalities; 3) local frequency extraction through partitioned phase-aware processing; and 4) a hierarchical superposition fusion (HSF) mechanism that iteratively integrates spatial and frequency information across levels using frequency modulation.

$$F_{\text{local}}^{\text{lidar}} = \mathcal{R}\left(\mathcal{C}_1\left(\mathcal{F}^{-1}\left(\text{Cat}\left(\mathbf{A}_{\text{lidar}}^{i}, e_i^{i\Phi_{\text{hsi}}}\right)\right)\right)\right) \tag{17}$$

$$F_l = \mathcal{R}\left(\mathcal{C}_1\left(F_{\text{local}}^{\text{hsi}} + F_{\text{local}}^{\text{lidar}}\right)\right). \tag{18}$$

where $F_{\text{local}}^{\text{hsi}}$ and $F_{\text{local}}^{\text{lidar}}$ denote the local frequency-domain features of HSI and LiDAR data, respectively.

*4) HSF Module:* To seamlessly integrate information from both the spatial and frequency domains and ensure feature consistency, we inject a tightly connected, hierarchical learning paradigm to achieve multilevel frequency-domain feature fusion. Specifically, for the features $F_s$, $F_g$, and $F_l$ extracted from the three FE blocks, we first perform frequency modulation on $F_g$ to further refine its characteristics. Subsequently, we overlay the refined global frequency feature $\hat{F}_g$ onto the spatial and local frequency features, continually applying frequency modulation. This process can be expressed as

$$\hat{F}_g = \mathcal{M}\left(F_g\right) \tag{19}$$

$$\hat{F}_s = \mathcal{M}\left(\hat{F}_g + F_s\right) \tag{20}$$

$$\hat{F}_l = \mathcal{M}\left(\hat{F}_s + F_l\right) \tag{21}$$

$$F_{\text{fused}} = \hat{F}_g + \hat{F}_s + \hat{F}_l \tag{22}$$

where $\mathcal{M}(\cdot)$ represents the frequency modulation operation and $F_{\text{fused}}$ denotes the final fused feature map incorporating spatial- and frequency-domain information.

The essence of frequency modulation lies in maintaining feature consistency. As illustrated in Fig. 4, this is achieved by removing irrelevant noise from the amplitude [48]. Specifically, we first transform the input features into the Fourier domain based on spatial dimensions and extract the corresponding amplitude spectrum $A_c$. Subsequently, we average the input features along the channel dimension to produce a channel-invariant feature $\mathbf{x}_{ci} \in \mathbb{R}^{H\times W\times 1}$. We then apply a Fourier transform to $\mathbf{x}_{ci}$, generating its amplitude $\mathbf{x}_{ci}$, which includes noise information independent of the channels.

Finally, we subtract $A_{ci}$ from $A_c$ to remove feature discrepancies. The result is then processed through a projection network comprising a global average pooling layer, two consecutive linear layers, and a sigmoid activation function to generate a conditioning vector, ensuring consistency across heterogeneous features.

### D. Learnable Voting Pre-Label Fusion

The outputs from the three modality-specific branches are first converted into class probability vectors, denoted as $t_1$–$t_3$. Each vector represents the predicted probability distribution over all classes for a given pixel. To fuse these predictions effectively, we assign a learnable weight to each branch: $U$, $M$, and $D$, corresponding to $t_1$–$t_3$, respectively. These weights are constrained such that $U + M + D = 1$, ensuring a normalized and balanced contribution from each branch during training. The final class prediction for each pixel is obtained by computing a weighted sum of the three class probability vectors. Specifically, for each class index $n$, the aggregated score $\theta(\text{pixel}, n)$ is computed as follows:

$$\theta(\text{pixel}, n) = U \cdot t_1(n) + M \cdot t_2(n) + D \cdot t_3(n). \tag{23}$$

The final predicted class for the pixel is determined by taking the class with the highest aggregated score

$$\text{Out}(\text{pixel}) = \arg\max_{n=1,\ldots,N} \theta(\text{pixel}, n) \tag{24}$$

where $N$ is the total number of classes and $\theta(\text{pixel}, n)$ denotes the fused classification confidence for class $n$. This learnable voting-based fusion mechanism enables the model to adaptively weight information from different branches according to their reliability and relevance. By leveraging complementary strengths across modalities, the model achieves more robust and generalizable classification, particularly in complex terrain scenarios.

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TABLE I
TRAINING AND TEST SAMPLES IN MUUFL, TRENTO, AUGSBURG, AND HOUSTON2013 DATASETS

| Class | MUUFL | | | Trento | | | Augsburg | | | Houston2013 | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | **Land Cover** | **Train** | **Test** | **Land Cover** | **Train** | **Test** | **Land Cover** | **Train** | **Test** | **Land Cover** | **Train** | **Test** |
| 01 | Trees Mostly | 465 | 22781 | Apple Trees | 41 | 3993 | Forest | 271 | 13236 | Healthy Grass | 63 | 1188 |
| 02 | Grass | 86 | 4184 | Buildings | 30 | 2873 | Residential Area | 607 | 29722 | Stressed Grass | 63 | 1191 |
| 03 | Mixed Ground Surface | 138 | 6744 | Ground | 5 | 474 | Industrial Area | 78 | 3773 | Synthetic Grass | 35 | 662 |
| 04 | Dirt and Sand | 37 | 1789 | Woods | 92 | 9031 | Low Plants | 538 | 26319 | Trees | 63 | 1181 |
| 05 | Road | 134 | 6553 | Vineyard | 106 | 10395 | Allotment | 12 | 563 | Soil | 63 | 1179 |
| 06 | Water | 10 | 456 | Roads | 32 | 3142 | Commercial Area | 33 | 1612 | Water | 17 | 308 |
| 07 | Buildings Shadow | 45 | 2188 | | | | Water | 31 | 1499 | Residential | 64 | 1204 |
| 08 | Buildings | 125 | 6115 | | | | | | | Commercial | 63 | 1181 |
| 09 | Sidewalk | 28 | 1357 | | | | | | | Road | 63 | 1189 |
| 10 | Yellow Curb | 4 | 179 | | | | | | | Highway | 62 | 1165 |
| 11 | Cloth Panels | 6 | 263 | | | | | | | Railway | 62 | 1173 |
| 12 | | | | | | | | | | Parking Lot1 | 62 | 1171 |
| 13 | | | | | | | | | | Parking Lot2 | 24 | 445 |
| 14 | | | | | | | | | | Tennis Court | 22 | 406 |
| 15 | | | | | | | | | | Running Track | 33 | 627 |
| Total | | 1078 | 52609 | | 306 | 29908 | | 1570 | 76724 | | 759 | 14270 |

## III. EXPERIMENTS

### A. Datasets Settings

Table I reports the individual categories and the number of training and testing samples for all used datasets.

**MUUFL Gulfport Dataset** was collected over Gulfport, Mississippi, USA, combining high-resolution hyperspectral and LiDAR data to support both urban and natural terrain analysis. Hyperspectral data were captured using the Reflective Optics System Imaging Spectrometer (ROSIS), spanning the spectral range of 0.375–1.05 $\mu$m and covering 64 bands within the visible and near-infrared spectrum. The spatial resolution of the hyperspectral imagery is 1 m/pixel, with LiDAR data providing a point spacing of approximately 1–2 cm, yielding detailed 3-D structural information. The dataset features 11 land cover classes and has a resolution size of 325 $\times$ 320 pixels.

**Trento Dataset** was captured over rural and agricultural areas in Trento, Italy. The HSIs were acquired using the AISA Eagle sensor, which covers the spectral range from 0.4 to 1.0 $\mu$m with 63 spectral bands. These images have a spatial resolution of 1 m/pixel, with a total image size of 166 $\times$ 600 pixels. LiDAR data were collected using the Optech ALTM 3100EA airborne LiDAR sensor. The dataset includes six land cover classes for classification purposes.

**Augsburg Dataset** was acquired over the city of Augsburg, Germany. The hyperspectral data were collected using the HySpex sensor, spanning from 0.4 to 2.5 $\mu$m across 160 bands, covering visible, near-infrared, and shortwave infrared regions. The spatial resolution of the HSIs is 0.5 m/pixel, with a resolution size of 332 $\times$ 485 pixels. The dataset also includes LiDAR data with a point spacing of 1–2 cm. It contains 78 294 pixels for classification, distributed across seven land cover classes.

**Houston2013 Dataset** was obtained using the ITRES CASI-1500 sensor, which primarily captured HSIs of the University of Houston campus and its surrounding urban areas. Operating in the spectral range of 0.38–1.05 $\mu$m, the sensor collects data across 144 spectral bands, facilitating detailed spectral analysis from visible to near-infrared wavelengths. In addition to the HSIs, the dataset also includes LiDAR data. Each hyperspectral image has a spatial resolution of 349 $\times$ 1905 pixels, and the dataset contains a total of 15 029 pixels for classification. The dataset is divided into 15 distinct land cover classes.

### B. Experimental Settings

*1) Performance Metrics:* To comprehensively evaluate the classification performance of the proposed framework and comparative models, we utilized four widely adopted metrics: overall accuracy (OA), average accuracy (AA), the Kappa coefficient ($k$), and per-class accuracy.

OA reflects the proportion of correctly classified samples across all categories, serving as an indicator of the model's general performance. AA calculates the mean accuracy of all classes, highlighting the model's ability to handle imbalanced datasets. $k$ measures the agreement between predicted and true labels, accounting for random chance, offering a robust evaluation of classification consistency. Per-class accuracy evaluates the classification accuracy for each category, revealing the model's capability to distinguish between different classes. Higher values for each of these metrics signify better classification performance, with greater precision and reliability.

*2) Experimental Environment Configuration:* All experiments, including those using the proposed method and other deep learning models, were conducted on a PyTorch platform hosted on an AMAX tower server. The server featured an Intel Xeon Silver 4210 processor running at 2.20 GHz, 256 GB of RAM, and an NVIDIA GeForce RTX 4090 GPU with 24 GB of VRAM. The Adam optimizer was chosen to train the network, with a batch size of 32 and a total of 100 epochs set for the training process.

### C. Parameter Exploration

To validate the optimal performance of our model, we systematically investigated several hyperparameters that could influence classification accuracy and the training process.

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These hyperparameters include the reduced spectral bands $b$, the size of the input data patch $s$, the learning rate $r$, and the scaling factor $m$ for the low-frequency mask.

*1) Reduced Spectral Band:* To investigate the effect of the reduced spectral band on classification performance, the proposed FHPF-Net was evaluated using datasets from MUUFL, Trento, Augsburg, and Houston2013. The spectral bands were reduced to dimensions $\{5, 10, 15, 20, 25, 30, 35, 40\}$ using PCA before being fed into the network. Fig. 5 illustrates the relationship between the number of bands and classification metrics, including OA, AA, and Kappa coefficient, across all four datasets.

The results demonstrate that as the number of spectral bands increases, the performance initially improves, reaching its peak when the dimensionality is set to 30. Beyond this point, further increases in dimensionality yield marginal or no improvement, while computational costs continue to rise. Therefore, considering both classification performance and efficiency, the reduced spectral band dimension of 30 was chosen as the optimal value for subsequent experiments.

*2) Patch Size:* To evaluate the impact of input patch size on classification performance, we conducted experiments across four datasets, varying the patch size within the range of $\{7 \times 7, 9 \times 9, 11 \times 11, 13 \times 13, 15 \times 15, 17 \times 17\}$.

As depicted in Fig. 6, classification performance improves as the patch size increases, reaching its peak at a size of $11 \times 11$. Beyond this point, performance either stagnates or slightly declines, which could be attributed to the inclusion of irrelevant spatial information in larger patches. Consequently, $11 \times 11$ is identified as the optimal patch size, striking a balance between accuracy and computational complexity.

*3) Learning Rate:* The learning rate was varied across $1e^{-5}$, $5e^{-5}$, $1e^{-4}$, $5e^{-4}$, $1e^{-3}$, $5e^{-3}$ to analyze its impact on the classification performance of our proposed FHPF-Net.

As depicted in Fig. 7, the results indicate that a learning rate of $5e^{-4}$ achieves the best performance on the MUUFL, Augsburg, and Houston2013 datasets, while the Trento dataset performs optimally at $1e^{-3}$. These findings suggest that although $5e^{-4}$ is a generally effective choice, dataset-specific tuning can further enhance classification accuracy.

*4) Scaling Factor for the Low-Frequency Mask:* The HLSSM module incorporates a scaling factor $m$, which determines the proportion of spectral dimensions assigned to the low-frequency mask, thereby isolating critical low-frequency components for analysis. To evaluate its influence on classification performance, $m$ was varied among $\{2, 4, 6, 8\}$.

As depicted in Fig. 8, the classification results demonstrate that $m = 4$ consistently yields the highest accuracy across all four datasets. This finding suggests that isolating one-fourth of the spectral dimensions effectively preserves essential low-frequency features while minimizing redundancy. Based on these results, $m = 4$ was adopted for subsequent experiments, underscoring its role in optimizing the balance between feature preservation and noise suppression.

### D. Classification Performance and Discussion

To comprehensively evaluate the performance of the proposed FHPF-Net, experiments were conducted on four benchmark datasets: MUUFL, Trento, Augsburg, and Houston2013. Two categories of methods were selected for comparison: 1) HSI-based classification methods, such as SVM [20], GSC-ViT [28], and MiM [49]; and 2) joint HSI and LiDAR classification methods, including MFTNet [50], CoupledCNN [51], CALC [52], HCTNet [35], M2FNet [53], HLMamba [54], and HFSNet [40]. To ensure a fair comparison, the parameters for all baseline methods were set to the optimal values reported in their respective papers. Table I provides a summary of the distribution of training and testing samples for each dataset. Both quantitative evaluations and visual assessments were performed to analyze the classification results in detail.

*1) Quantitative Assessment and Analysis:* Table II presents the classification accuracy on the MUUFL Gulfport dataset. Our FHPF-Net demonstrates superior performance across various landform categories, particularly excelling in scenarios with complex terrain, showcasing its robust capability in joint spectral and spatial feature learning.

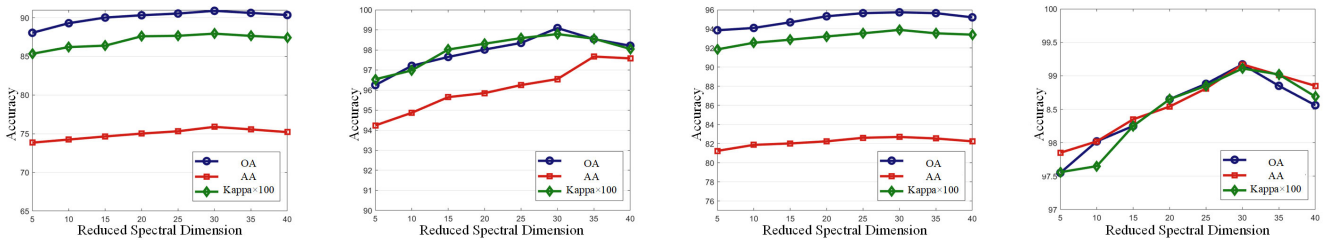
Compared with HSI-based classification methods, our FHPF-Net achieves leading accuracy in multiple categories, notably in categories 1, 3, and 8. Specifically, in category 8 (Buildings), our FHPF-Net attains an outstanding classification accuracy of 97.37%, significantly surpassing GSC-ViT and MiM, which achieved 94.32% and 95.52%, respectively. Additionally, our FHPF-Net exhibits superior overall performance, achieving the highest OA of 90.89%, AA of 75.89%, and Kappa coefficient of 87.94, outperforming traditional classifiers such as RF and SVM by a clear margin.

Compared with HSI and LiDAR joint classification methods, our FHPF-Net further amplifies its advantages. It achieves state-of-the-art accuracy in categories such as 1, 3, and 6, where complex spectral–spatial interactions are prominent. In category 6, our FHPF-Net achieves an exceptional accuracy of 94.01%, far exceeding CoupledCNN (75.92%), M2FNet (86.55%), and HFSNet(90.06%). Similarly, for category 3, FHPF-Net achieves 86.72%, outperforming CALC (84.67%), HCTNet (82.33%), and HFSNet (83.39%). The comprehensive utilization of multimodal data allows FHPF-Net to capture intricate patterns effectively, contributing to its leading OA, AA, and Kappa values, surpassing advanced methods such as MFTNet, HLMamba, M2FNet, and HFSNet. These results highlight FHPF-Net's ability to harness the synergistic benefits of HSI and LiDAR data, outperforming state-of-the-art methods in both single-modal and multimodal scenarios.

Table III reports the classification performance of various methods on the Trento dataset and demonstrates a consistent advantage in both scenarios, achieving state-of-the-art results in several landform categories and metrics. Across all metrics, FHPF-Net exhibits superior results, with an OA of 99.09%, an AA of 96.55%, and a Kappa coefficient of 98.79, achieving state-of-the-art performance.

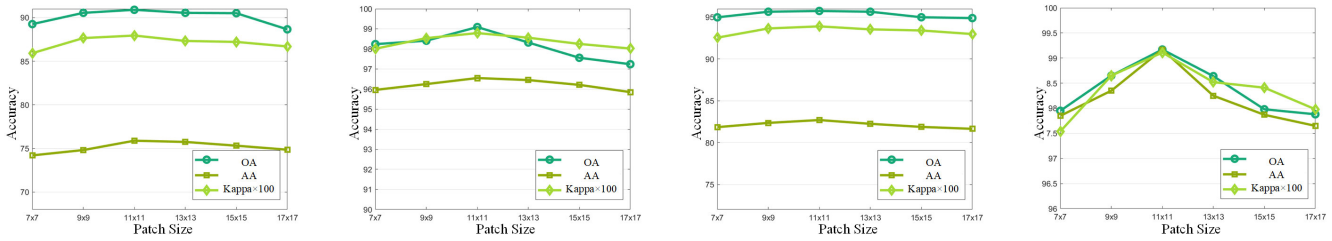
Our proposed FHPF-Net surpassed HSI-based classification methods in all key indicators and achieved the highest OA of 97.72%, exceeding MiM (94.67%), and GSC-ViT (94.43%) while significantly outperforming the traditional machine learning model like SVM (94.96%). In individual categories, FHPF-Net excels, particularly in categories

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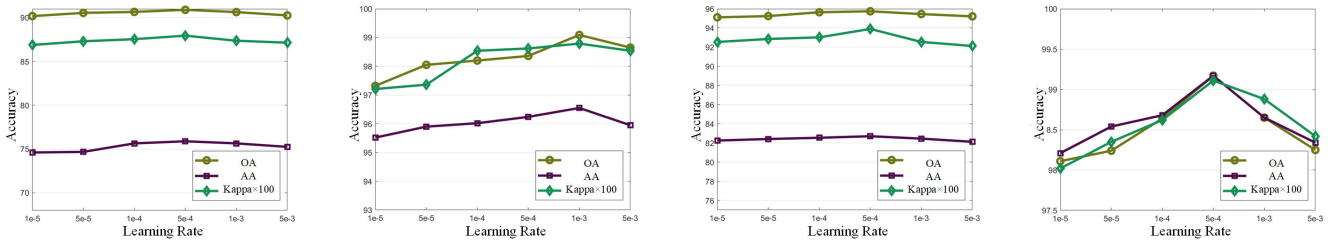

(a) (b) (c) (d)

Fig. 5. Impact of reduced spectral dimensionality on classification metrics: OA, AA, and Kappa coefficient. (a) MUUFL. (b) Trento. (c) Augsburg. (d) Houston2013.



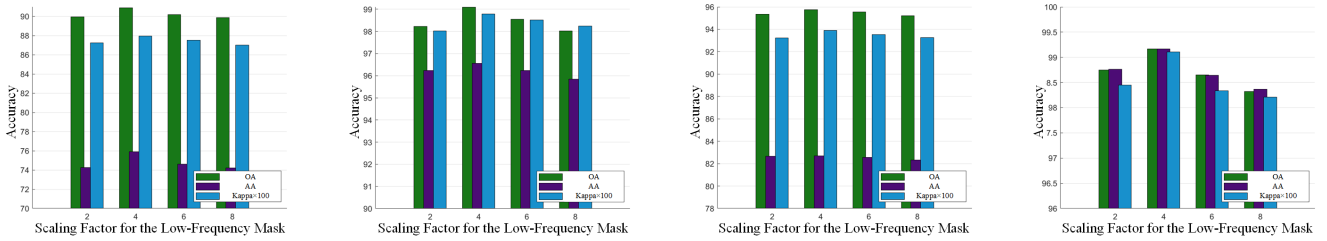

(a) (b) (c) (d)

Fig. 6. Impact of patch size on classification metrics: OA, AA, and Kappa coefficient. (a) MUUFL. (b) Trento. (c) Augsburg. (d) Houston2013.




(a) (b) (c) (d)

Fig. 7. Impact of learning rate on classification metrics: OA, AA, and Kappa coefficient. (a) MUUFL. (b) Trento. (c) Augsburg. (d) Houston2013.




(a) (b) (c) (d)

Fig. 8. Impact of scaling factor for the low-frequency mask on classification metrics: OA, AA, and Kappa coefficient. (a) MUUFL. (b) Trento. (c) Augsburg. (d) Houston2013.

2 and 6, achieving accuracies of 95.48% and 97.27%, respectively, demonstrating robust spectral–spatial learning capabilities. When both HSI and LiDAR data are combined, our FHPF-Net further enhances its performance. It achieves leading accuracy in categories such as 1 (Apple Trees) and 2 (Buildings), with category-specific accuracies of 99.67% and 98.83%, respectively. These results outperform other advanced multimodal methods, including M2FNet (98.80% OA), HLMamba (98.74% OA), and HFSNet (98.81% OA). Notably, in category 4 (Woods), FHPF-Net achieves perfect accuracy (100.0%), tying with HLMamba while slightly surpassing MFTNet (99.99%). The ability to integrate HSI and LiDAR data allows FHPF-Net to effectively capture complex multimodal patterns, solidifying its performance advantage.

Table IV presents the classification results for the Augsburg dataset using various methods. The results demonstrate that methods combining HSI and LiDAR generally achieve superior performance compared to those relying solely on HSI, particularly in categories 3, 5, and 6.

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TABLE II

CLASSIFICATION ACCURACY OF COMPARISON EXPERIMENT IN MUUFL GULFPORT USING VARIOUS METHODS. NOS. 1–11 ARE THE CLASSIFICATION ACCURACIES FOR EACH LANDFORM CATEGORY

| No. | HSI-based classification method | | | HSI and LiDAR joint classification methods | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | SVM [20] | GSC-ViT [28] | MiM[49] | MFTNet [50] | CoupledCNN [51] | CALC [52] | HCTnet [35] | M2FNet [53] | HLMamba [54] | HFSNet [40] | **Ours** |
| 1 | 95.68±0.69 | 94.18±0.9 | 92.17±3.76 | 94.44±1.23 | 96.05±0.02 | 94.13±1.6 | 94.62±0.6 | 94.38±1.12 | 96.25±1.0 | 96.4±0.75 | **96.52±0.35** |
| 2 | **91.75±1.76** | 88.54±1.02 | 84.77±1.58 | 86.98±1.67 | 89.2±0.24 | 88.06±2.31 | 89.05±0.58 | 76.77±5.02 | 86.31±0.84 | 82.21±1.76 | 85.46±1.59 |
| 3 | 78.84±2.16 | 83.55±3.04 | 83.57±8.77 | 81.38±1.72 | 85.26±1.1 | 84.67±1.82 | 82.33±4.61 | 85.71±3.28 | 84.57±1.69 | 83.93±2.44 | **86.72±0.8** |
| 4 | 91.08±2.55 | 94.11±1.85 | 90.89±1.24 | 89.92±4.14 | 92.27±1.11 | 93.09±2.0 | **94.52±1.95** | 83.16±17.56 | 91.71±1.41 | 91.0±1.08 | 89.25±0.5 |
| 5 | 92.8±0.75 | 92.13±1.28 | 93.67±1.22 | 91.7±2.87 | 89.65±0.38 | 91.92±0.39 | 93.48±2.77 | **94.41±0.39** | 87.78±1.95 | 90.92±0.75 | 90.82±0.71 |
| 6 | 63.2±4.62 | 67.68±15.66 | 66.88±9.01 | 69.2±24.26 | 75.92±7.95 | 75.99±2.11 | 90.6±4.85 | 86.55±4.05 | 74.55±12.17 | 90.06±0.99 | **94.01±1.86** |
| 7 | 80.53±3.17 | 80.61±1.32 | 77.45±6.01 | 85.16±3.3 | 82.6±2.18 | **87.55±2.01** | 71.85±11.06 | 84.17±1.84 | 78.73±2.11 | 80.47±2.88 | 78.61±2.11 |
| 8 | 94.96±1.06 | 94.32±0.11 | 95.52±2.89 | 94.22±2.22 | 94.86±0.76 | 94.63±1.73 | 91.25±6.91 | 93.02±3.78 | 92.6±3.75 | 94.65±1.13 | **97.37±0.01** |
| 9 | 2.11±0.93 | 7.29±2.03 | 17.36±5.98 | 26.22±7.24 | 1.89±0.33 | 13.07±7.05 | 17.02±7.84 | 20.87±4.85 | 14.67±6.02 | 38.86±3.72 | **41.78±0.52** |
| 10 | 7.26±3.9 | 10.06±2.99 | 13.59±7.58 | 16.95±12.45 | 4.47±0.46 | 13.59±1.47 | **28.49±10.09** | 6.7±1.64 | 17.88±0.46 | 7.08±1.32 | 10.24±1.47 |
| 11 | 44.49±0.31 | **83.02±2.07** | 57.92±6.69 | 64.26±1.35 | 71.48±8.09 | 71.48±5.75 | 77.82±0.95 | 54.88±11.75 | 79.47±1.08 | 54.5±1.77 | 64.01±5.45 |
| OA (%) | 88.71±0.60 | 88.74±0.12 | 87.81±1.18 | 88.86±0.98 | 89.45±0.31 | 89.26±0.38 | 88.76±0.93 | 88.45±0.95 | 88.95±1.00 | 89.89±0.48 | **90.89±0.18** |
| AA (%) | 67.52±1.19 | 72.32±1.90 | 70.35±0.76 | 72.77±2.65 | 71.24±1.73 | 73.47±0.54 | 75.55±1.52 | 70.97±1.24 | 73.14±2.32 | 73.64±0.02 | **75.89±0.36** |
| $k$×100 | 85.05±0.77 | 85.11±0.10 | 83.95±1.47 | 85.30±1.30 | 86.00±0.43 | 85.86±0.44 | 85.18±1.24 | 84.74±1.25 | 85.28±1.37 | 86.61±0.63 | **87.94±0.23** |

TABLE III

CLASSIFICATION ACCURACY OF COMPARISON EXPERIMENT IN TRENTO USING VARIOUS METHODS. NOS. 1–6 ARE THE CLASSIFICATION ACCURACIES FOR EACH LANDFORM CATEGORY

| No. | HSI-based classification method | | | HSI and LiDAR joint classification methods | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | SVM [20] | GSC-ViT [28] | MiM[49] | MFTNet [50] | CoupledCNN [51] | CALC [52] | HCTnet [35] | M2FNet [53] | HLMamba [54] | HFSNet [40] | **Ours** |
| 1 | 97.67±1.29 | 96.58±0.94 | 97.28±0.9 | 95.42±1.49 | 98.87±0.5 | 99.55±0.23 | 98.02±0.58 | 99.13±0.5 | **99.87±0.06** | 99.74±0.19 | 99.67±0.09 |
| 2 | 76.66±4.97 | 70.49±2.85 | 95.48±3.93 | 89.47±0.98 | 94.84±1.62 | 94.08±3.16 | 89.72±3.68 | 97.84±0.59 | 95.77±0.89 | 95.2±3.23 | **98.83±0.61** |
| 3 | 57.35±17.07 | 82.98±3.52 | 79.06±5.32 | **93.82±3.13** | 84.21±8.1 | 78.88±12.23 | 86.22±3.85 | 76.02±7.52 | 78.13±0.65 | 81.36±20.27 | 85.74±5.09 |
| 4 | 99.58±0.41 | 99.76±0.16 | 99.94±0.05 | 99.49±0.72 | 99.96±0.02 | 99.82±0.03 | 99.89±0.1 | 99.99±0.01 | **100.0±0.0** | **100.0±0.0** | **100.0±0.0** |
| 5 | 99.96±0.04 | 99.24±0.28 | 97.57±1.72 | 99.96±0.04 | 97.55±2.09 | 99.72±0.23 | 99.91±0.12 | 99.73±0.11 | 99.97±0.05 | 99.89±0.08 | **99.99±0.02** |
| 6 | 84.12±5.22 | 86.31±0.64 | 97.27±0.41 | 97.59±2.13 | 93.77±1.52 | **97.99±0.8** | 94.97±2.8 | 96.2±2.42 | 95.41±1.98 | 96.58±1.32 | 95.06±0.91 |
| OA (%) | 94.96±0.23 | 94.67±0.46 | 97.72±0.34 | 97.86±0.38 | 97.59±0.87 | 98.67±0.31 | 97.94±0.44 | 98.80±0.32 | 98.74±0.13 | 98.81±0.81 | **99.09±0.10** |
| AA (%) | 85.89±2.13 | 89.23±0.69 | 94.43±1.23 | 95.96±0.78 | 94.87±1.41 | 95.01±1.89 | 94.79±0.87 | 94.82±1.11 | 94.86±0.30 | 95.46±4.17 | **96.55±0.87** |
| $k$×100 | 93.26±0.31 | 92.87±0.61 | 96.96±0.45 | 97.14±0.51 | 96.79±1.15 | 98.23±0.41 | 97.24±0.59 | 98.40±0.43 | 98.31±0.17 | 98.41±1.08 | **98.79±0.14** |

TABLE IV

CLASSIFICATION ACCURACY OF COMPARISON EXPERIMENT IN AUGSBURG USING VARIOUS METHODS. NOS. 1–7 ARE THE CLASSIFICATION ACCURACIES FOR EACH LANDFORM CATEGORY

| No. | HSI-based classification method | | | HSI and LiDAR joint classification methods | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | SVM [20] | GSC-ViT [28] | MiM[49] | MFTNet [50] | CoupledCNN [51] | CALC [52] | HCTnet [35] | M2FNet [53] | HLMamba [54] | HFSNet [40] | **Ours** |
| 1 | 97.49±0.37 | 97.33±0.8 | 98.36±0.42 | 97.89±0.26 | 98.63±0.05 | 98.06±1.02 | 95.57±3.21 | 97.99±0.16 | **98.92±0.32** | 97.13±1.03 | 97.92±0.6 |
| 2 | **98.94±0.13** | 98.61±0.22 | 98.92±0.26 | 97.54±1.11 | 98.09±0.06 | 96.05±2.74 | 97.81±0.8 | 97.03±1.42 | 94.06±2.02 | 98.14±0.57 | 97.62±0.48 |
| 3 | 70.76±3.64 | 84.94±2.97 | 79.14±4.16 | 81.64±2.39 | 85.21±1.81 | 70.1±4.1 | 79.24±6.75 | 80.33±11.93 | 83.7±14.95 | 74.8±13.43 | **86.57±2.47** |
| 4 | 98.3±0.24 | 98.04±0.79 | 97.79±0.48 | **98.94±0.37** | 98.38±0.23 | 97.76±0.55 | 97.37±1.05 | 98.72±0.3 | 98.75±0.28 | 98.25±0.42 | 98.53±0.38 |
| 5 | 4.56±1.03 | 57.89±3.58 | 64.32±3.03 | 65.08±7.49 | 61.63±3.52 | 46.0±22.73 | 73.42±16.18 | 63.9±9.72 | 71.37±18.15 | 64.95±6.12 | **80.48±5.46** |
| 6 | 25.98±7.46 | 49.63±2.46 | 31.03±9.36 | 39.83±10.02 | 23.18±0.33 | **61.41±10.25** | 44.84±14.09 | 23.57±6.98 | 22.26±19.37 | 53.54±9.83 | 55.69±2.63 |
| 7 | 58.13±1.12 | **65.67±2.23** | 63.15±3.94 | 63.78±5.17 | 57.37±0.38 | 64.06±1.6 | 61.42±2.8 | 65.11±3.64 | 61.06±6.39 | 63.26±5.11 | 62.05±3.98 |
| OA (%) | 94.06±0.07 | 95.55±0.17 | 95.09±0.16 | 95.19±0.12 | 95.01±0.13 | 93.99±1.43 | 94.35±0.83 | 94.53±0.18 | 93.67±1.13 | 94.99±0.31 | **95.74±0.28** |
| AA (%) | 64.88±0.71 | 78.87±0.60 | 76.10±0.80 | 77.81±3.26 | 74.64±0.39 | 76.21±3.10 | 78.52±1.75 | 75.24±2.37 | 75.73±2.42 | 78.58±1.96 | **82.70±0.64** |
| $k$×100 | 91.39±0.11 | 93.60±0.24 | 92.93±0.21 | 93.10±0.20 | 92.82±0.19 | 91.44±1.97 | 91.92±1.15 | 92.15±0.28 | 91.01±1.57 | 92.80±0.46 | **93.90±0.40** |

In categories 3 and 5, our method exhibits remarkable improvements compared to most other approaches, achieving the highest accuracy of 86.57% and 80.48%, respectively. These gains reflect the capability of our method to effectively leverage the complementary information from HSI and LiDAR data for complex landform classification. For category 6, while CoupledCNN achieves the highest accuracy, our method also demonstrates competitive performance, significantly surpassing most other methods.

In terms of OA, our approach achieves 95.74%, which is the best among all methods. Additionally, we observe improvements in AA and the Kappa coefficient, indicating robust performance across various classes. This highlights the adaptability and effectiveness of our method in integrating multimodal data for remote sensing classification tasks.

Table V presents the classification results for the Houston2013 dataset using various methods. As with the previous datasets, we compare methods using only HSI input and those utilizing both HSI and LiDAR data. The results indicate that methods using HSI and LiDAR inputs generally achieve better performance than those relying solely on HSI, especially in categories 1, 4, and 15. Our method achieves the highest accuracy in most categories, including category 1, with an accuracy of 98.99%, and category 6, where it also reaches 100.0%.

In terms of OA, our approach achieves 99.17%, surpassing all other methods. Additionally, we observe significant

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TABLE V
CLASSIFICATION ACCURACY OF COMPARISON EXPERIMENT IN THE HOUSTON2013 DATASET USING VARIOUS METHODS. NOS. 1–15 ARE THE CLASSIFICATION ACCURACIES FOR EACH LANDFORM CATEGORY

| No. | HSI-based classification method | | | HSI and LiDAR joint classification methods | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | SVM [20] | GSC-ViT [28] | MiM[49] | MFTNet [50] | CoupledCNN [51] | CALC [52] | HCTnet [35] | M2FNet [53] | HLMamba [54] | HFSNet [40] | **Ours** |
| 1 | 97.79±0.7 | 94.57±0.5 | 98.12±1.27 | 98.03±0.51 | 85.41±0.34 | 97.63±0.49 | 98.35±0.28 | 97.63±0.21 | 96.8±1.28 | 98.26±0.67 | **98.99±0.45** |
| 2 | 99.43±0.33 | 99.48±0.23 | 98.07±1.25 | 99.77±0.15 | 98.2±1.81 | 99.75±0.2 | 99.06±0.53 | 99.48±0.14 | 99.64±0.14 | 98.07±1.03 | **99.8±0.09** |
| 3 | 97.84±1.49 | **100.0±0.0** | **100.0±0.0** | 99.73±0.28 | 99.5±0.26 | 99.6±0.14 | **100.0±0.0** | **100.0±0.0** | 99.9±0.07 | 99.04±0.61 | 98.13±0.56 |
| 4 | 98.33±0.68 | 99.81±0.04 | 92.64±0.16 | 99.75±0.14 | **99.97±0.04** | 94.2±1.23 | 99.61±0.36 | 99.22±0.58 | 96.34±0.58 | 98.73±1.09 | 99.44±0.49 |
| 5 | **100.0±0.0** | 99.75±0.35 | **100.0±0.0** | 99.98±0.03 | 99.92±0.12 | **100.0±0.0** | **100.0±0.0** | **100.0±0.0** | **100.0±0.0** | 99.97±0.04 | 99.88±0.17 |
| 6 | 98.32±0.54 | 96.85±0.93 | 85.49±1.36 | 99.16±0.91 | 97.94±0.15 | 92.97±2.29 | 98.51±0.91 | 97.51±0.55 | 86.58±0.15 | 100.0±0.0 | **100.0±0.0** |
| 7 | 88.46±1.35 | 91.5±3.15 | 93.45±2.76 | 99.19±0.44 | 96.28±0.69 | 97.28±1.0 | 99.34±0.33 | 93.31±2.31 | 94.86±2.61 | 99.17±0.83 | **99.68±0.47** |
| 8 | 90.46±4.4 | 93.1±1.65 | 86.04±1.55 | 90.55±2.78 | 94.31±1.13 | 97.66±1.46 | 96.41±1.76 | 96.2±0.99 | **99.44±0.34** | 97.91±0.24 | 97.39±0.88 |
| 9 | 81.93±3.67 | 76.93±1.94 | 90.06±2.06 | 96.89±0.99 | 96.44±2.0 | 95.93±2.1 | **99.01±0.63** | 94.98±0.67 | 93.02±1.45 | 93.66±1.96 | 97.49±0.88 |
| 10 | 99.53±0.41 | 99.02±0.68 | 97.3±1.98 | 99.71±0.14 | 99.61±0.56 | **100.0±0.0** | 99.9±0.14 | 99.49±0.12 | 99.83±0.12 | 98.48±0.45 | **100.0±0.0** |
| 11 | 98.03±0.44 | 99.61±0.14 | 97.37±1.84 | 97.53±0.35 | 95.75±2.82 | 95.84±1.43 | 99.74±0.25 | 98.18±0.16 | 99.66±0.07 | 98.35±1.13 | **99.81±0.06** |
| 12 | 88.91±0.99 | 98.06±1.12 | 99.53±0.2 | 90.63±0.27 | **99.66±0.0** | 98.92±0.28 | 94.57±0.07 | 97.81±1.38 | 98.55±0.14 | 99.26±0.47 | 99.64±0.03 |
| 13 | 78.05±7.3 | 90.88±1.43 | 90.08±0.9 | 96.99±0.58 | 91.99±1.33 | 93.48±0.18 | 98.92±0.33 | 92.66±1.04 | **99.48±0.28** | 98.43±1.15 | 97.21±0.69 |
| 14 | **100.0±0.0** | **100.0±0.0** | 99.92±0.11 | **100.0±0.0** | **100.0±0.0** | **100.0±0.0** | **100.0±0.0** | **100.0±0.0** | **100.0±0.0** | **100.0±0.0** | **100.0±0.0** |
| 15 | **100.0±0.0** | 99.95±0.07 | 99.74±0.37 | **100.0±0.0** | **100.0±0.0** | **100.0±0.0** | **100.0±0.0** | 99.68±0.34 | **100.0±0.0** | **100.0±0.0** | **100.0±0.0** |
| OA (%) | 94.43±0.21 | 95.63±0.12 | 95.42±0.39 | 97.55±0.22 | 96.79±0.18 | 97.74±0.25 | 98.78±0.16 | 97.73±0.18 | 97.87±0.43 | 98.40±0.39 | **99.17±0.16** |
| AA (%) | 94.47±0.29 | 95.97±0.19 | 95.19±0.32 | 97.86±0.18 | 97.00±0.13 | 97.55±0.09 | 98.89±0.11 | 97.74±0.07 | 97.61±0.36 | 98.62±0.37 | **99.17±0.13** |
| $k$×100 | 93.98±0.22 | 95.28±0.13 | 95.05±0.42 | 97.35±0.24 | 96.52±0.19 | 97.55±0.27 | 98.68±0.18 | 97.54±0.20 | 97.70±0.47 | 98.27±0.42 | **99.11±0.17** |

improvements in AA and the Kappa coefficient, highlighting the robustness of our method in utilizing multimodal data for classification. These results demonstrate the effectiveness of integrating HSI and LiDAR data to enhance classification performance in complex urban environments.

*2) Visual Assessment and Analysis:* In this section, we provide a visual assessment and comparison of the classification results for the four datasets: MUUFL, Trento, Augsburg, and Houston2013. Each dataset presents its unique challenges, but our proposed FHPF-Net shows promising results across the board.

As shown in Fig. 9, the MUUFL dataset, known for its complex urban environments and varying landforms, demonstrates a clear advantage when using FHPF-Net. Compared to the traditional method, which tends to show scattered misclassifications, FHPF-Net offers a much cleaner classification output, with distinct boundaries between landform categories. Methods like GSC-ViT and MFTNet show good performance but are still prone to slight misclassifications along edge regions, especially in densely built-up areas. FHPF-Net, however, achieves superior accuracy and offers a more precise delineation of the landform classes.

As shown in Fig. 10, the Trento dataset consists of mixed urban and agricultural landscapes, and the difference in performance becomes more pronounced. While SVM produces classification maps with significant noise, especially in the transition zones between agricultural and urban areas, our FHPF-Net provides a more accurate representation of these transitions. The performance of models like CoupledCNN, CALC, and HFSNet is similar, but they still struggle with finer details and edge detection. Our proposed FHPF-Net consistently performs better, capturing the finer features and nuances of the landform categories.

As shown in Fig. 11, the Augsburg dataset includes a variety of land types and shows that our FHPF-Net outperforms other models, particularly when distinguishing between subtle landform classes such as forests and agricultural fields. SVM is unable to effectively differentiate between these classes in certain regions, whereas FHPF-Net provides highly accurate and visually consistent classification maps. Methods like M2FNet, HCTNet, and HFSNet show improvements but still lag behind our FHPF-Net in terms of classification clarity and boundary precision.

As shown in Fig. 12, the Houston2013 dataset encompasses highly urbanized regions, presenting a significant challenge for accurate classification. Our proposed FHPF-Net stands out due to its ability to delineate terrain with exceptional accuracy. Traditional methods, such as SVM and RF, struggle in these complex regions, often failing to capture intricate spatial and spectral details. More advanced models, such as MFTNet and HFSNet, provide reasonable results; however, they frequently exhibit blurred category boundaries, particularly in densely urbanized areas. In contrast, our FHPF-Net demonstrates the capability to produce sharp and well-defined terrain boundaries. This is attributed to its effective integration of spectral, spatial, and elevation features, enabling superior classification performance even in highly challenging urban settings. Consequently, our FHPF-Net delivers significantly improved classification outcomes compared to both traditional and state-of-the-art methods.

Overall, the visual comparison reveals that our FHPF-Net outperforms the other methods across all datasets, providing sharper boundaries, fewer misclassifications, and a more accurate depiction of landform classes.

### *E. Comparison With Frequency-Based Methods*

To comprehensively evaluate the effectiveness of our method, we compare it with three state-of-the-art frequency-domain-based approaches: FDNet [41], MCFNet [43], and CHNet [38]. The comparison is conducted on four benchmark datasets: MUUFL Gulfport, Trento, Augsburg, and Houston2013. Results are summarized in Table VI and Fig. 13.

*1) Classification Accuracy Analysis:* Compared with the other frequency-based models, our method achieves

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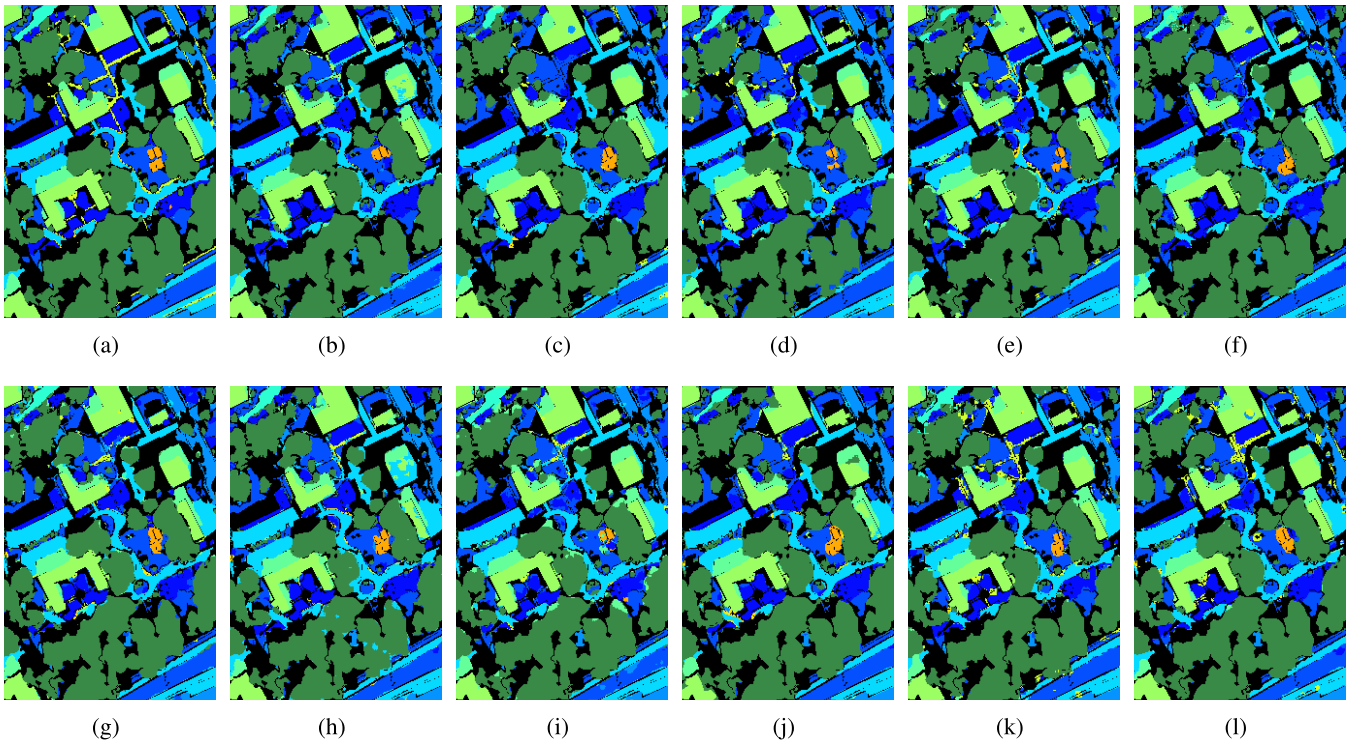



Fig. 9. Maps depicting the classification of MUUFL using various methods. (a) Ground truth. (b) SVM. (c) GSC-ViT. (d) MiM. (e) MFTNet. (f) CoupledCNN. (g) CALC. (h) HCTnet. (i) M2FNet. (j) HLMamba. (k) HFSNet. (l) FHPF-Net.

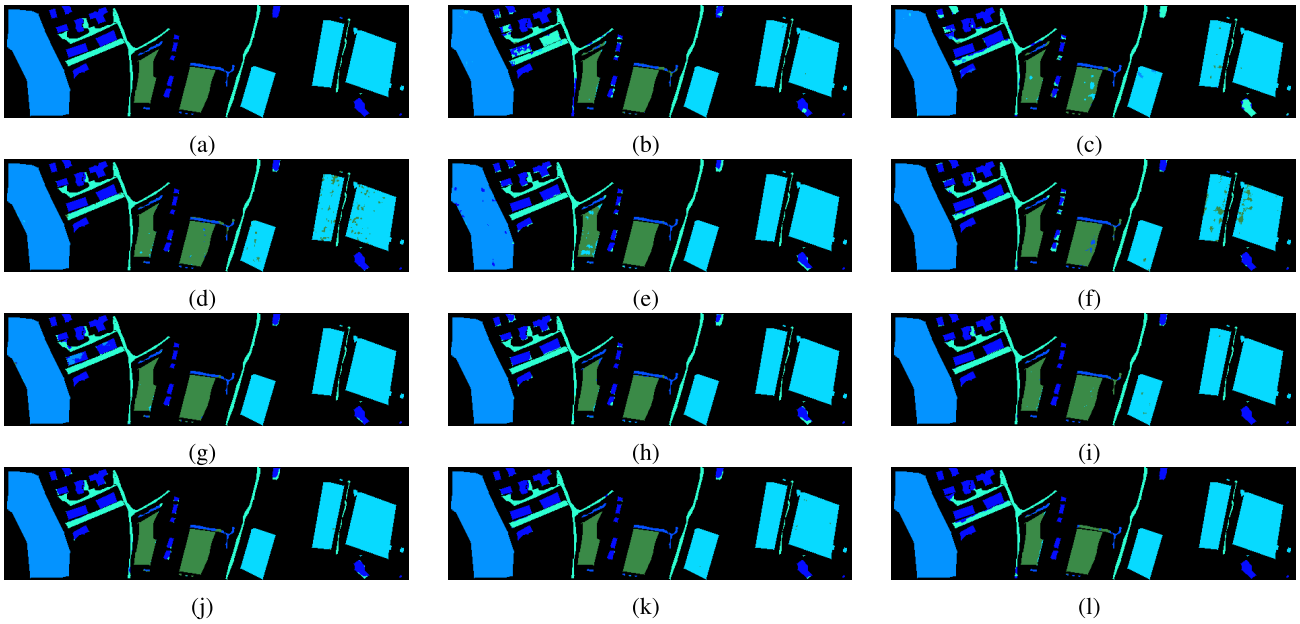



Fig. 10. Maps depicting the classification of Trento using various methods. (a) Ground truth. (b) SVM. (c) GSC-ViT. (d) MiM. (e) MFTNet. (f) CoupledCNN. (g) CALC. (h) HCTnet. (i) M2FNet. (j) HLMamba. (k) HFSNet. (l) FHPF-Net.

consistently better performance across all datasets and metrics. For example, on the MUUFL dataset, our method surpasses FDNet and MCFNet by over 0.6% in OA and more than 3% in AA. This improvement stems from the joint modeling of global and local frequency features: FDNet and MCFNet primarily capture global frequency characteristics via DWT or Fourier-based operations but lack sufficient attention to localized spectral variations or fail to align them effectively with spatial features. In contrast, our method integrates global frequency representations to enhance structural awareness, while local frequency features capture subtle texture and region-level discriminative details. Their hierarchical fusion enables more robust semantic discrimination. On high-resolution datasets like Trento and Houston2013, the advantage becomes more pronounced. Our model's OA reaches 99.09% and 99.17%, respectively, which demonstrates its strong adaptability to both

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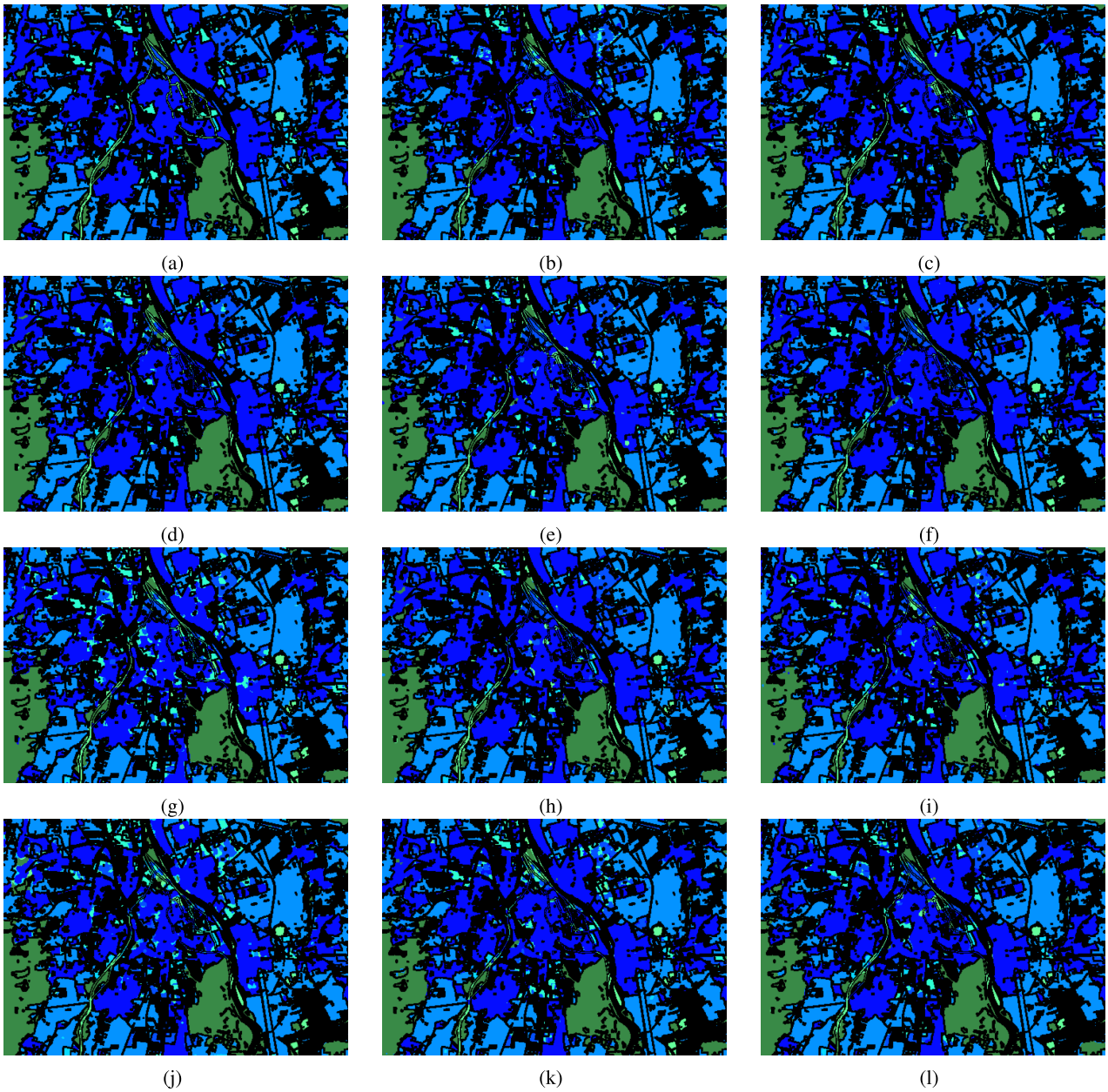



Fig. 11. Maps depicting the classification of Augsburg using various methods. (a) Ground truth. (b) SVM. (c) GSC-ViT. (d) MiM. (e) MFTNet. (f) CoupledCNN. (g) CALC. (h) HCTnet. (i) M2FNet. (j) HLMamba. (k) HFSNet. (l) FHPF-Net.

TABLE VI

CLASSIFICATION ACCURACY OF COMPARATIVE EXPERIMENTS CONDUCTED ON DIFFERENT DATASETS USING FREQUENCY-BASED METHODS

| Indicators | MUUFL Gulfport | | | | Trento | | | |
|---|---|---|---|---|---|---|---|---|
| | FDNet [41] | MCFNet [43] | CHNet [38] | Ours | FDNet [41] | MCFNet [43] | CHNet [38] | Ours |
| OA (%) | 90.27±0.24 | 89.76±0.90 | 88.28±0.31 | **90.89±0.18** | 98.59±0.15 | 98.81±0.23 | 95.65±0.23 | **99.09±0.10** |
| AA (%) | 72.43±0.51 | 72.84±2.24 | 70.73±1.86 | **75.89±0.36** | 94.20±0.81 | 94.97±1.67 | 79.95±0.16 | **96.55±0.87** |
| $k$×100 | 87.07±0.34 | 86.42±1.17 | 84.39±0.41 | **87.94±0.23** | 98.12±0.20 | 98.40±0.30 | 94.15±0.30 | **98.79±0.14** |
| Indicators | Augsburg | | | | Houston2013 | | | |
| | FDNet [41] | MCFNet [43] | CHNet [38] | Ours | FDNet [41] | MCFNet [43] | CHNet [38] | Ours |
| OA (%) | 95.18±0.10 | 94.19±0.25 | 94.30±0.08 | **95.74±0.28** | 96.25±0.60 | 93.96±0.33 | 97.36±0.04 | **99.17±0.16** |
| AA (%) | 73.31±0.99 | 70.52±1.96 | 71.67±0.84 | **82.70±0.64** | 96.88±0.44 | 94.46±0.28 | 97.50±0.05 | **99.17±0.13** |
| $k$×100 | 93.07±0.15 | 91.63±0.39 | 91.75±0.11 | **93.90±0.40** | 95.94±0.65 | 93.46±0.36 | 97.14±0.05 | **99.11±0.17** |

fine-grained local variations and large-scale spatial structures, benefiting from frequency–space interactions.

*2) Local-Visualization Analysis:* As shown in Fig. 13, frequency-based baselines (FDNet, MCFNet, and CHNet) tend

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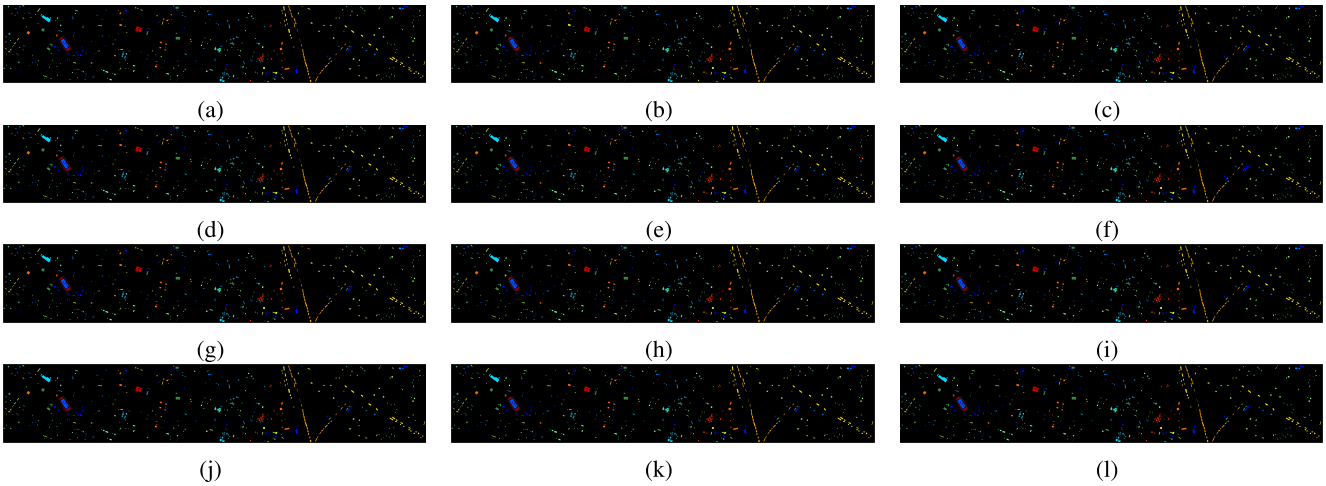



Fig. 12. Maps depicting the classification of Houston2013 using various methods. (a) Ground truth. (b) SVM. (c) GSC-ViT. (d) MiM. (e) MFTNet. (f) CoupledCNN. (g) CALC. (h) HCTnet. (i) M2FNet. (j) HLMamba. (k) HFSNet. (l) FHPF-Net.

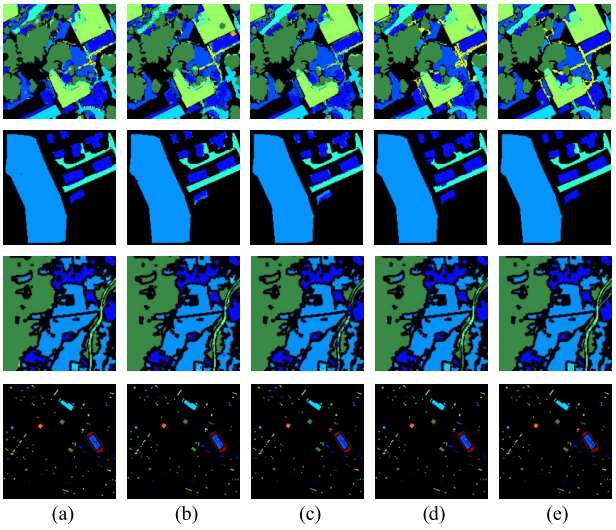



Fig. 13. Comparison of the frequency-based method with the proposed local visualization of classification maps on the four datasets used in this article. (a) FDNet. (b) MCFNet. (c) CHNet. (d) Ours. (e) Ground truth.

to suffer from fragmented prediction and spatial inconsistency, especially in spectrally similar or boundary regions. This reflects their limited ability to balance spectral resolution and spatial smoothness. Our method alleviates this problem through the hierarchical superimposed fusion of frequency and spatial domains, which enhances the interclass boundary clarity and intraclass consistency. Particularly on complex scenes such as MUUFL and Augsburg, our results show significantly fewer noisy predictions and better alignment with ground truth. These observations confirm that our frequency-aware fusion strategy is not only numerically superior but also visually more reliable, providing more structured and semantically meaningful outputs.

## IV. Ablation Analysis

### A. Exploration of Ablation Results for Various Data Modalities

The ablation experiments for various data modalities were conducted on four datasets: MUUFL, Trento, Augsburg, and Houston2013, with the sample distribution shown in Table I. In these experiments, we evaluated the performance of our method by comparing the classification results using only HSI data, only LiDAR data, and the combination of both HSI and LiDAR data.

The experimental results are shown in Table VII. When only HSI data are used, the model achieves relatively good performance, with a notable improvement in OA, AA, and Kappa across most datasets. For instance, on the MUUFL dataset, the OA reaches 87.56%, and on Trento, it achieves 95.65%. However, when only LiDAR data are used, the performance drops considerably, particularly in terms of both OA and AA. For example, on MUUFL, the OA falls to 70.46%, and on Trento, it is only 75.65%. This highlights the inherent limitations of using LiDAR alone, as it lacks the spectral richness of HSI data, which is essential for capturing fine details like texture and material properties.

In contrast, when both HSI and LiDAR data are used in conjunction, the model shows substantial improvements in all evaluation metrics. For instance, on the MUUFL dataset, the OA increases to 90.89%, AA increases to 75.89%, and the Kappa score increases to 87.94%. Similarly, on Trento, OA increases to 99.09% and AA increases to 96.55%. This demonstrates the synergy between the two modalities, where HSI provides the detailed spectral information, and LiDAR contributes valuable spatial and structural features. The fusion of these two data sources enables the model to achieve more accurate and reliable classification results across different environments and datasets.

### B. Ablation Results for Different Components

To comprehensively evaluate the contribution of each core component in FHPF-Net, we conducted ablation experiments across four representative datasets: MUUFL, Trento, Augsburg, and Houston2013. The proposed architecture is composed of four key modules: FE, HLSSM, HSMIF, and LVPF. As shown in Table VIII, we assess the performance using the OA metric by selectively removing each module.

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TABLE VII
RESULTS OF ABLATION EXPERIMENTS FOR VARIOUS DATA MODALITIES

| Cases | MUUFL | | | Trento | | | Augsburg | | | Houston2013 | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | OA(%) | AA(%) | $k \times 100$ | OA(%) | AA(%) | $k \times 100$ | OA(%) | AA(%) | $k \times 100$ | OA(%) | AA(%) | $k \times 100$ |
| Only HSI | 87.56 | 72.35 | 82.35 | 95.65 | 92.63 | 94.75 | 90.35 | 77.41 | 88.39 | 94.25 | 93.87 | 93.95 |
| Only LiDAR | 70.46 | 58.42 | 67.62 | 75.65 | 72.63 | 73.92 | 67.62 | 57.62 | 62.42 | 59.66 | 62.35 | 58.21 |
| HSI+LiDAR | **90.89** | **75.89** | **87.94** | **99.09** | **96.55** | **98.79** | **95.74** | **82.7** | **93.9** | **99.17** | **99.17** | **99.11** |

TABLE VIII
RESULTS OF ABLATION EXPERIMENTS

| Cases | Component | | | | MUUFL | Trento | Augsburg | Houston2013 |
|---|---|---|---|---|---|---|---|---|
| | FE | HLSSM | HSMIF | LVPF | OA(%) | OA(%) | OA(%) | OA(%) |
| 1 | × | ✓ | ✓ | ✓ | 89.88 | 96.51 | 94.43 | 97.47 |
| 2 | ✓ | × | ✓ | ✓ | 89.46 | 96.54 | 94.52 | 97.54 |
| 3 | ✓ | ✓ | × | ✓ | 89.57 | 96.79 | 94.64 | 98.17 |
| 4 | ✓ | ✓ | ✓ | × | 90.55 | 98.67 | 95.27 | 98.82 |
| 5 | ✓ | ✓ | ✓ | ✓ | **90.89** | **99.09** | **95.74** | **99.17** |

In Case 1, the FE module is removed, meaning that no preliminary spatial–spectral features are extracted before subsequent processing. This results in a noticeable OA drop across all datasets (e.g., from 90.89% to 89.88% on MUUFL), indicating that foundational feature representations are essential for downstream modules to operate effectively.

In Case 2, excluding the HLSSM module leads to further OA reductions. For example, the OA on the MUUFL and Augsburg datasets decreases to 89.46% and 94.52%, respectively. This suggests that the HLSSM module, which facilitates the separation of high- and low-frequency spectral patterns, plays a critical role in isolating complementary information and enhancing discriminative power.

Case 3 omits the HSMIF module, removing the hierarchical multidomain fusion strategy. This leads to a performance decline on all datasets, especially on Houston2013 (from 99.17% to 98.17%), highlighting the module's effectiveness in integrating spatial- and frequency-domain features from heterogeneous modalities.

In the absence of the LVPF module, the network cannot dynamically assign weights to different modalities, leading to suboptimal fusion of heterogeneous data. While the performance remains relatively high, the lack of adaptivity in modality fusion slightly hinders the model's classification robustness. This suggests that LVPF is essential for optimizing the final decision-making process based on the relative importance of each feature set.

In conclusion, each module in FHPF-Net contributes uniquely to the model's performance. The FE module provides essential foundational features, HLSSM enhances feature quality by focusing on high- and low-frequency details, HSMIF ensures effective multidomain integration, and LVPF enables adaptive fusion. The combined effect of these components results in a significant improvement in classification accuracy and robustness.

### C. Ablation Results for Feature Components in the HSMIF

To investigate the individual contributions of spatial, global frequency, and local frequency features within the HSMIF module, we conducted a series of ablation experiments, and the results are summarized in Table IX. In Case 1, only spatial information is preserved, yielding an OA of 88.75%, which demonstrates the fundamental role of spatial cues in HSI-LiDAR joint classification. When global frequency features are introduced alongside spatial features (Case 2), the overall performance improves to 89.56% OA, suggesting that the global spectral structure provides beneficial complementary information that enhances the model's understanding of scene-level semantics. Similarly, incorporating local frequency features with spatial cues (Case 3) also leads to performance gains, indicating that local spectral variations contribute significantly to fine-grained feature discrimination.

TABLE IX
RESULTS OF ABLATION EXPERIMENTS FOR FEATURE COMPONENTS IN THE HSMIF

| Cases | Feature Component | | | Indicators | | |
|---|---|---|---|---|---|---|
| | Spatial | Global Frequency | Local Frequency | OA(%) | AA(%) | $k \times 100$ |
| 1 | ✓ | × | × | 88.75 | 73.45 | 85.79 |
| 2 | ✓ | ✓ | × | 89.56 | 74.11 | 86.66 |
| 3 | ✓ | × | ✓ | 89.49 | 73.82 | 86.51 |
| 4 | × | ✓ | ✓ | 88.68 | 74.72 | 85.21 |
| 5 | ✓ | ✓ | ✓ | **90.89** | **75.89** | **87.94** |

In Case 4, where spatial features are removed but both frequency components are retained, the model suffers a noticeable drop in OA to 88.68%. This highlights that although frequency-domain features carry rich spectral context, the absence of spatial structure undermines pixel-level localization, which is critical in remote sensing tasks. Finally, Case 5 integrates all three components: spatial, global frequency, and local frequency, achieving the best performance with an OA of 90.89%, an AA of 75.89%, and a Kappa of 87.94. These results validate the hierarchical and superimposed design of the HSMIF module, showing that multidomain feature fusion yields synergistic benefits. By effectively capturing spatial geometry, global semantics, and local texture cues, the full module enhances both classification accuracy and robustness, particularly in complex, heterogeneous terrain environments.

### D. Computational Cost Comparison Experiment

To comprehensively evaluate the tradeoff between performance and computational burden, we compare our proposed method with four recent deep learning approaches: MFTNet, HCTnet, M2FNet, and CHNet across multiple indicators: OA, training time, test time, model parameters, and floating-point operations (FLOPs).

As shown in Table X, our method achieves the highest OA, surpassing all baselines by a notable margin, especially

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TABLE X
COMPUTATIONAL COST COMPARISON EXPERIMENTAL RESULTS (OPTIMAL RESULTS ARE BOLDED)

| Methods | Accuracy and Efficiency Indicators | | | | |
|---|---|---|---|---|---|
| | OA(%) | Train Time (s) | Test Time (s) | Parameters (M) | FLOPs (G) |
| MFTNet | 88.86±0.98 | 175.64 | 18.77 | 0.31 | 1.63 |
| HCTnet | 88.76±0.98 | **167.37** | 16.57 | 0.45 | **0.69** |
| M2FNet | 88.45±0.95 | 197.54 | 18.24 | **0.26** | 5.80 |
| CHNet | 88.28±0.31 | 326.43 | 22.63 | 20.77 | 10.29 |
| Ours | **90.89±0.18** | 217.57 | **15.22** | 1.15 | 1.43 |

compared to the low-FLOPs methods like MFTNet and HCTnet. Although HCTnet has the lowest FLOPs, its OA falls short, indicating that blind parameter compression may compromise discriminative capacity. In contrast, CHNet requires significantly more computation but still lags behind in accuracy (88.28%), suggesting an inefficient feature design. Our method maintains a highly competitive FLOPs value and a compact model size while delivering superior accuracy. This highlights the effectiveness of the frequency-aware modules, which enhance spectral–spatial representation with minimal overhead. In terms of speed, our method yields the fastest inference time (15.22 s) while maintaining a reasonable training duration (217.57 s), close to other lightweight methods. This makes it well-suited for real-time or resource-constrained applications.

In summary, our method achieves an optimal balance between precision and efficiency, outperforming lightweight models in accuracy and heavy models in computational cost, demonstrating its strong potential for scalable deployment.

## V. CONCLUSION

In this study, we developed the FHPF-Net to tackle the challenges of joint classification for HSI and LiDAR remote sensing data. Unlike conventional approaches that rely heavily on spatial-domain feature fusion, our method integrates frequency-domain analysis to enhance the representation and fusion of spectral–spatial–elevation features. To address the need for effective frequency-domain representation, we propose the HLSSM module. It separates high- and low-frequency components, enabling the extraction of fine-grained textures and broader structural patterns from HSI and LiDAR data while preserving their unique spectral and elevation features. Furthermore, we proposed the HSMIF module to achieve seamless integration of heterogeneous features. By leveraging a tightly connected hierarchical stacking learning paradigm, the HSMIF module facilitates the comprehensive fusion of spatial- and frequency-domain information, ensuring feature consistency and complementarity across modalities. Finally, we propose the LVPF strategy, which efficiently combines the outputs from multiple branches to improve classification accuracy and bolster the model's robustness. The experimental results demonstrate the superiority of FHPF-Net over existing methods, achieving significant improvements in classification accuracy across various remote sensing benchmarks.

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